RISK AND RETURN

3

THE 21ST CENTURY: A TIME OF FINANCIAL CRISIS AND RECOVERY FOR EQUITIES

The global financial crisis, market turmoil and falling equity values reminded investors that investing in equities often involves taking on significant levels of risk. We should not forget how large the falls in equity values were in late 2008 and early 2009. Credit Suisse reported that the developed World Index fell by 55% which represented a loss of R150 trillion or R150 000 for every man, woman and child in the developed world[1]. This is serious, but we should also consider equity returns over the longer term and that additional risks should be rewarded with higher returns. The developed world index for the ten years to 2010 offered investors a return that was close to zero. It began with the crash of technology and telecommunication equity values in the USA, in 2000, when the NASDAQ index lost about 80% of its value.

Emerging markets performed strongly and offered very positive returns for investors up to 2011. South Africa is an emerging market and experienced positive gains in equity values from 2001 to 2013. The question is whether the higher returns in emerging markets came with higher risks.

In 2013, 2014 and 2015, emerging markets experienced a significant amount of volatility and capital outflows which were matched with falls in currency values. In the USA, equity values increased significantly from 2010 due to very low interest rates, quantitative easing (the printing of money), and economic recovery matched with significant corporate tax cuts. The S&P500 went from about 1100 in January 2010 to 3329 in January 2020. Then Covid hit global markets in March 2020 and the S&P500 fell to 2305. However, significant government stimulus, ultralow interest rates and quantitative easing resulted in global share markets recovering strongly. The return of inflation in 2021 and 2022 resulted in the rise in global interest rates and this resulted in a significant fall in the S&P500 and the MSCI World Index[2] by December 2022. The NASDAQ fell by 33%. However, the fall in the inflation rate in 2023 particularly in the USA and the advances in AI has resulted in a strong recovery

in the values of the major IT companies in the USA. The MSCI World Index was close to its all-time high in July 2023 reaching 300 as compared to 100 in July 2008. At the same time, the MSCI Emerging markets index had only risen to 144 in July 2023 from 100 in July 2008. In terms of the USA and the developed economies, the market was again reaching new highs, although the distribution of such gains, were mostly limited to the very large IT companies. Is overconfidence in equity markets going to lead to a fall in equity values? Low volatility and rising equity values may lead investors to take on greater risks until we reach a Minsky moment when "stability begets instability"[3]. Is it possible that behavioural finance can explain market movements and may it guide us to be calm when others are diving into pools of "irrational exuberance" or despair?

In this chapter, we firstly analyse risk in terms of operating and financial leverage. The chapter explains expected returns, the normal distribution and measures of volatility such as standard deviation. The chapter concludes with the analysis of risk and return in financial markets.

LEARNING OBJECTIVES

After working through this chapter, you should be able to:

- Distinguish between business risk and financial risk.
- Calculate indicators of **returns** based on earnings before interest and tax (EBIT), percentage return to shareholders, expected return based on probabilities, arithmetic average returns, compound annual returns and total shareholder returns. Page 3-2
- Explain and calculate indicators of **risk** such as degree of operating leverage, degree of financial leverage, variance of returns, standard deviation of returns, co-efficient of variation, the *z* score, covariance and correlation, and Value at Risk (VaR).
- Understand the history of equity returns and bond returns.
- Define risk-adjusted measures of performance.
- Explain the behavioural finance view of risk and return.

INTRODUCTION

Financial management is about making investment and financing decisions. Such decisions involve risk. What is risk? How do we determine required returns? It is understandable that an investment's risk will affect the required return. The higher the risk, the higher the required return. So, what kinds of risks do companies face? When a company builds a new manufacturing plant, there is the risk that the product will not sell sufficient units for the company to breakeven. Further, the company may not be able to achieve the expected price due to unexpected competitive pricing decisions. A high level of operating leverage may mean that a company's earnings are significantly affected by a small change in sales. A mining company such as Gold Fields may find that the initial capital cost and the ongoing costs of operating a gold mine may be significantly higher than expected at the time of the investment. A company borrowing funds may be subject to interest rate hikes which may result in reduced cash flows. Yet, companies may experience substantial improvements in operating returns due to unexpected price increases. For example, South32's investment in the Mozal smelter was undertaken when the price of aluminium was about $1 100 per ton. In 2006, the price was over $2 700 per ton. In 2008 the aluminium price fell to about $1 200 during the economic crisis. After recovering to reach about $2 000 in 2014, the price then collapsed in 2015 to under $1 500. However, by June 2018 the price of aluminium had recovered to $2 300 per ton. In late August 2023, the aluminium price was $2 156 per ton. Aluminium is a major component of numerous products and this price volatility creates risks for manufacturers. Risk is often measured in terms of the volatility of expected returns and the potential for loss. In order to make investment decisions we need to measure and adjust for risk.

Few questions about future events can be answered with certainty, because we don't know whether these decisions are good or bad until future events either confirm or repudiate the expectations on which the decisions were based. Money is invested by financial managers in assets which, it is forecast, will generate income and increase the value of the firm. Similarly, investors make investment decisions based on their expectations, and only in the future will it be possible to determine whether these investments were profitable. Evidently, not every investment decision will be profitable, because no one can consistently make accurate predictions about the future.

Making predictions is clearly hazardous because you will sometimes be wrong. Methods must be sought to keep the margin of error as low as possible. Information

about the past must rank as a significant input into predictions about the future. However, if it is known that the past is unlikely to be replicated in the future, information about past events is of little value and must be replaced with realistic estimates of future expectations. This chapter suggests methods by which expected return can be measured. It discusses the uncertainty that surrounds such expected returns – i.e. the risk that the returns may not be achieved – and measures this risk. This requires the application of certain basic statistical techniques.

Once we determine how to measure an investment's expected return and its associated risk, we compare investments with different risk and return profiles. We thereby lay the foundation for the study of portfolio theory in Chapter 4.

The historical evidence of share returns will be examined in the last section of this chapter. We evaluate the international and South African evidence of whether the higher risk of investing in equities was rewarded with higher returns. Did equities offer returns that were higher than bonds? Page 3-3

Investment risk can be measured by comparing the risk of one investment with that of the market as a whole. This risk comparison is denoted by a firm's beta which measures the relative volatility of a share to the market. If a firm's beta is 1.2, then this means that the firm's shares are more volatile than the market. If the market goes up by 5%, then we would expect the share price of the firm to go up by 6% (5% × 1.2). The market is assigned a beta score of 1.0 and every share has a score that can be measured against this norm. We write about this measure of risk in Chapter 4.

1 THE CONCEPT OF RISK

The word *risk* is usually used in the context of a potential hazard or the possibility of loss resulting from a given action. In financial management, the term indicates that there is an expectation that the actual outcome of a project may differ from the expected outcome. The magnitude of the possible difference reflects the magnitude of the risk. Risk may also be positive as you may receive more than you expected from your investment.

The terms *risk* and *uncertainty* are often used interchangeably. There is, however, a formal difference between these two terms. *Uncertainty* implies either that all the alternative possible outcomes cannot be identified or that no probability can be attached to the alternative possible outcomes. *Risk* implies that it is possible to attach

probabilities to identified expected outcomes. This strict distinction between the two terms is, however, not necessary for our purposes.

From the viewpoint of an investor, investment opportunities can be broadly classified into two categories:

- *Investment opportunities with certain outcomes* (no expectation that the actual outcome will differ from the expected outcome); and
- *Investment opportunities with uncertain outcomes* (some probability that the actual outcome will differ from the expected outcome).

An example of a certain outcome is an investment in 8% government bonds, maturing in ten years. This may be regarded by some to have a measure of uncertainty should the government not honour its obligation, but insolvency of the country implies that all other investments are at least as badly affected. The South African government, with its power to manage the money supply and to impose taxes, is accepted to be the ultimate benchmark of risklessness in the economy. Examples of investments with risky outcomes are the purchase of shares on the JSE or the purchase of fixed property with a view to deriving capital gains.

A pertinent question is: why would an investor prefer a risky investment to one that is risk-free? The answer is that it offers a higher expected return. However, this does not fully answer the question why some investors prefer risky investments while others avoid risk. The preference for risk which some investors exhibit is generally accepted to be a function of the utility which an individual derives from making the investment. Because individuals differ both in their needs and in their personality traits, it follows that there will always be a spectrum of investors, some of whom are more prone to looking for high-risk investments with high expected returns, while others will maximise their utility through accepting lower expected returns with concomitant lower risk.

Risk is also a function of age. Older investors approaching retirement do not readily embrace risk as they do not have the ability to replace investment losses whereas the young are more prone to embrace risk. This is why financial planners need to consider the age and thus the risk profile of their clients when compiling appropriate investment portfolios.

Turning specifically to investment in the shares of listed companies, there are two major areas of risk which will be considered by potential shareholders: *business risk*

and *financial risk*. Page 3-4

Business risk

Business risk results from the nature of the business itself. It includes all the uncertainty that surrounds the industry in which the business operates. This is reflected in the variability of sales and the structure of costs. The variability of sales results from factors such as the likelihood of increased competition, the availability of substitute products, and the effect of recession. Rapid changes in technology and market shifts are also factors that may lead to increased business risk. The structure of costs depends on the relationship between fixed and variable costs, illustrated in Example 3.1. The effect of business risk on financial performance is measured by the variability of earnings before interest and taxes (EBIT).

Example 3.1: Cost-volume-profit analysis

Leverage Ltd has the following budgeted information:

- Fixed costs (F) R10 million per year;
- Variable costs (VC) R400 per unit;
- Selling price (S) R1 000 per unit; and
- Expected demand 30 000 units (minimum).

Figure 3.1 illustrates these facts graphically. It is apparent that sales revenue increases in proportion to demand. Fixed costs remain constant within a relevant range regardless of the sales volume, while variable costs commence at zero and increase in proportion to units sold.

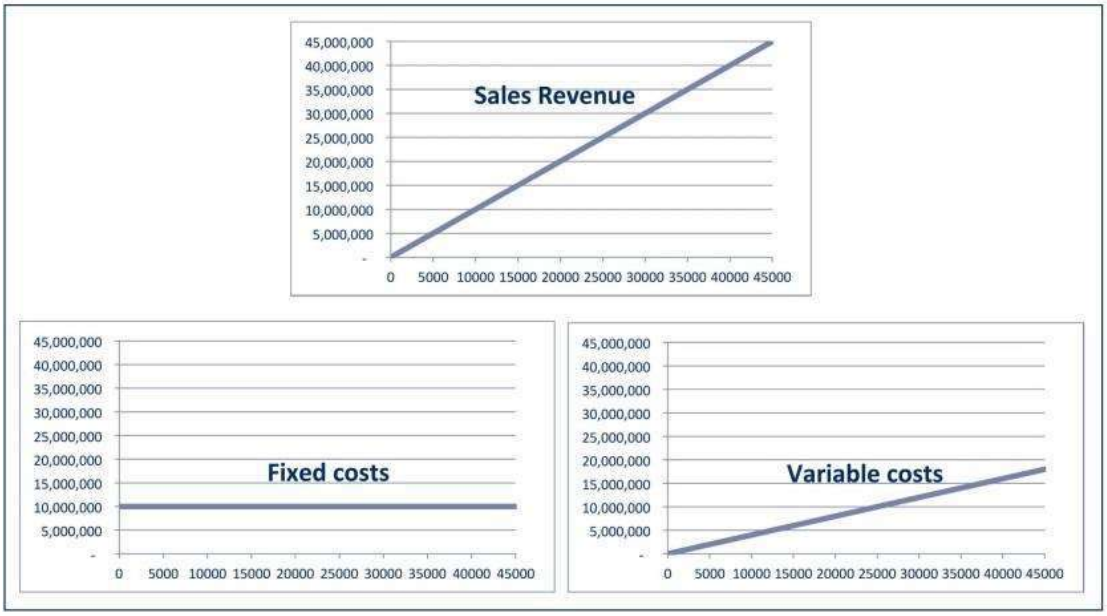


Figure 3.1 Behaviour of costs and revenues

In order to match sales revenue against total cost of sales, fixed costs (such as rent, depreciation, and administrative overheads) and variable costs (such as materials and direct labour) must be added together. The point at which sales revenue equals total cost is the volume of sales required in order to break even, that is, to make neither a profit nor a loss. This is illustrated in Figure 3.2. Page 3-5



Figure 3.2 Break-even volume

It is apparent from the graph that around 16 600 units must be sold in order to break even. If demand falls below 16 600, a loss will be incurred. Note that each time a unit

is sold, it makes a contribution of R600 toward the fixed cost (R1 000 – R400). No profit is earned until all fixed costs have been covered. The break-even point, that is, to make neither a profit nor a loss, can be precisely calculated using a simple formula, as follows:

$$\text{Break-even units} = \frac{\text{fixed costs}}{\text{contribution per unit}} \qquad \textit{(Formula 3.1)}$$

$$= \frac{F}{S - VC}$$

$$= \frac{10\,000\,000}{1\,000 - 400}$$

$$= 16\,667 \text{ units}$$

Once the break-even number of units has been sold, each additional unit sold will add its contribution (R1 000 – R400) to the profit. Table 3.1 indicates the effect on EBIT of different sales volumes.

Table 3.1 Effect of volume on EBIT of Leverage Ltd.

No. of units	30,000	40,000	% change
	Rm	Rm	
Sales (R1 000 per unit)	30.0	40.0	33%
Variable costs (R400 per unit)	12.0	16.0	33%
Contribution	18.0	24.0	33%
Fixed costs	10.0	10.0	0%
EBIT	8.0	14.0	75%

Page 3-6 It can be seen that, while sales increase by $33\frac{1}{3}\%$ from 30 000 units to 40 000 units, EBIT has increased by 75% from R8m to R14m. This results from the leverage effects of fixed costs. The degree of operating leverage obtained at 30 000 units can be expressed as follows:

$$\text{DOL} = \frac{\text{contribution}}{\text{EBIT}} \qquad \textit{(Formula 3.2)}$$

$$= \frac{S - VC}{S - VC - F}$$

In the case of Leverage Ltd, this can be calculated as:

$$\text{DOL at 30 000 units} = \frac{30\text{m} - 12\text{m}}{8\text{m}}$$

$$= 2.25$$

This means, for example, that an increase in sales contribution of 10% will lead to an increase of 10% × 2.25, that is 22.5% in EBIT. This can be checked against the figures produced in Table 3.1. An increase in sales of $33\frac{1}{3}\%$led to an increase in EBIT of 75% $(33\frac{1}{3}\% \times 2.25)$.Assume, now, that Leverage Ltd decides to install machinery which will increase the fixed costs by R5m and reduce the variable costs by R150 per unit to R250 per unit. The resulting graphic representation is illustrated in Figure 3.3.

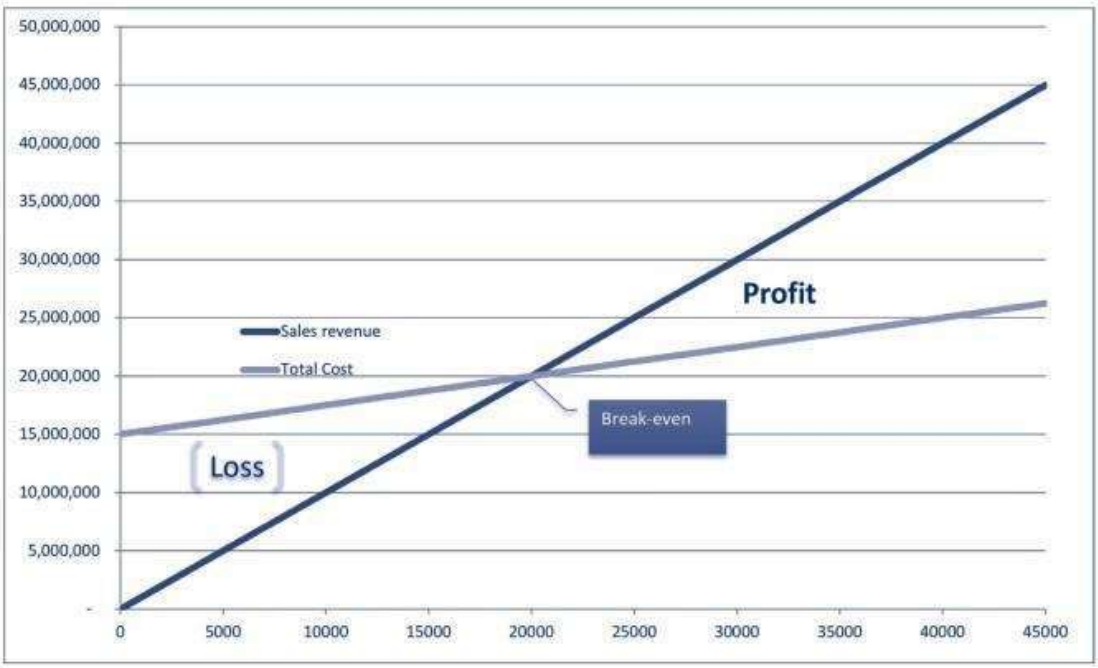


Figure 3.3 Change in fixed and variable costs

Page 3-7 From Figure 3.3 it is clear that the break-even point has shifted. It is determined using Formula 3.1:

$$\text{Break-even units} = \frac{15\,000\,000}{1\,000 - 250}$$
$$= 20\,000 \text{ units}$$

The degree of operating leverage will also change if we use the results illustrated below:

Table 3.2 Degree of operating leverage

No. of units	30,000	40,000	% change
	Rm	Rm	
Sales (R1 000 per unit)	30.0	40.0	33%
Variable costs (R250 per unit)	7.5	10.0	33%
Contribution	22.5	30.0	33%
Fixed costs	15.0	15.0	0%

No. of units	30,000	40,000	% change
EBIT	7.5	15.0	100%

The EBIT at 30 000 units is lower than it was previously, while at 40 000 units it is higher. This results from the higher fixed costs and lower variable costs. The DOL at 30 000 units is calculated as follows:

$$\text{DOL at 30 000 units} = \frac{30\text{m} - 7.5\text{m}}{7.5\text{m}}$$
$$= 3$$

It is confirmed by the fact that an increase of $33\frac{1}{3}\%$ in sales resulted in an increase of 100% $(33\frac{1}{3}\% \times 3)$ in EBIT.

Which cost structure is preferable? There is no correct answer to this question. The original cost structure is less risky as the break-even number of units of 16 667 is easier to attain than the 20 000 required by the second cost structure. However, as sales increase, a higher EBIT is obtained using the second cost structure. This is a classical risk/return issue. The riskier option offers greater potential losses if sales volume is low, but greater potential profits when sales volume is high. The total business risk is therefore a function both of sales and costs.

It is worth noting that managers do not have total control over the DOL. Often the nature of the industry prescribes the extent to which the business must incur fixed costs.

Financial risk

Financial risk results from the practice of financing a part of the firm's assets with interestbearing debt, with a view to increasing the ultimate return to the ordinary shareholders. Interest must be paid regardless of the performance of the firm. As a result, a firm which is liable for fixed interest payments is exposed to a risk of default which is not faced by a firm financed exclusively with shareholders' funds.

When the company is experiencing boom sales, the return on assets is likely to be considerably higher than the cost of the debt. As a result, positive financial leverage is experienced, which enhances the return on equity. However, in recessionary times the opposite effect is experienced, resulting in the possibility of defaulting on creditors. The company without debt is not subject to this risk, as all losses are fully absorbed by the shareholders.

The degree of financial leverage (DFL) arises only after the EBIT has been calculated. It is identical in concept to the DOL, as it increases fixed costs, being the fixed cost of interest which must be paid. The formula for determining the DFL is as follows:

$$DFL = \frac{EBIT}{EBIT - I} \qquad \textit{(Formula 3.3)}$$

where: EBIT = earnings before interest and tax

I = total interest expense on all debt

Total company risk

Operating leverage and financial leverage work together to create what is referred to as the *degree of combined leverage* (DCL). This results from the product of the degree of operating leverage and the degree of financial leverage and may be expressed as follows:

$$DCL = \frac{S - VC}{S - VC - F - I} \qquad \textit{(Formula 3.4)}$$

Stated more simply:

$$DCL = \frac{\text{Contribution}}{\text{Net income before taxation}}$$

DCL can also be stated as: DCL = DOL × DFL

Example 3.2:

Table 3.3 shows abbreviated contribution income statements of two companies.

Table 3.3 Income statements

	Leverhi	Gearlow
	Rm	Rm
Sales	81.0	89.1
Less: Variable operating costs	14.4	54.9
	66.6	34.2
Less: Fixed costs	33.3	6.3
Net income before interest	33.3	27.9
Interest	6.3	0.9
Net Income	27.0	27.0

Find the degree of operating leverage, the degree of financial leverage, and the degree of total leverage for each company. Using the appropriate formulae, the following results are obtained: Page 3-9

Table 3.4 Degree of combined leverage

			Leverhi	Gearlow
DOL =	$\frac{\text{S-VC}}{\text{S-VC-F}}$	=	$\frac{66.6}{33.3}$	$\frac{34.2}{27.9}$
		=	2.0000	1.2258
DFL =	$\frac{\text{EBIT}}{\text{EBIT - I}}$	=	$\frac{33.3}{27.0}$	$\frac{27.9}{27.0}$
		=	1.2333	1.0333
DCL =	DOL x DFL	=	2.4667	1.2667
or				
DCL =	$\frac{\text{S-VC}}{\text{S-VC-F-I}}$	=	$\frac{66.6}{27.0}$	$\frac{34.2}{27.0}$
		=	2.4667	1.2667

Although both companies have similar sales revenues, it emerges that the net income of Leverhi is considerably more sensitive to a decline in sales revenue. For example, if sales revenue declines by 20%, the net income of Leverhi will fall to R13.7m while the net income of Gearlow will only fall to R20.2m. The converse is also apparent in that an increase in sales revenue will benefit Leverhi considerably more than Gearlow. Again, this reflects the typical risk and return relationship. Leverhi is classified as a more risky business operation because of the greater potential loss in times of economic adversity. However, it has a greater potential for profit should sales revenue increase.

The use of DCL as a means of assessing risk is useful for the management of a company, particularly when budgets are being prepared or capital expenditure is being considered. Investors do not normally have access to this type of information. However, investors do understand that companies that operate in such sectors as the airline, oil, aluminium, chemical, shipping, steel, and manufacturing and hotel sectors will be subject to relatively high levels of operating leverage. These are capital intensive industries requiring significant investments in airplanes, property, ships, and plant and equipment. High levels of operating leverage may be accentuated by high levels of financial leverage. Companies that operate in the retail sector such as Pick n Pay and Foschini may have low levels of operating leverage. Retailers will however

often have fixed commitments in relation to long-term leases. Investors use mainly stock market indicators to assess risk.

2 MEASURING EXPECTED RETURN AND RISK

An investor is able to measure past returns from investment in a single share with relative ease. The two components of return are the dividends received and the capital or price appreciation of the share, expressed as follows:

$$R_s = \frac{P_1 - P_0 + D}{P_0} \qquad \textit{(Formula 3.5)}$$

where: R_s = return to shareholder

P_0 = share price at the beginning of the period

P_1 = share price at the end of the period

D = dividends received during the period Page 3-10

Depending on the purpose of the calculation, the period may be a week, a month, a year, or any other period. For the purpose of comparison, annual returns are most useful.

Note that the capital appreciation ($P_1 - P_0$) is included regardless of whether or not the shares were sold and the capital appreciation realised.

Like all decisions, the investment decision depends upon forecasting future events. Past returns may be a valuable input used in forecasting expected future returns. If, however, information is available which indicates that conditions in the future will differ from those in the past, such information must be taken into account when determining the expected return.

Measuring the expected return on a single share

Expected returns on an investment with an uncertain outcome cannot be accurately quantified. However, if probabilities can be assigned to a number of alternative outcomes, it is possible to statistically determine the expected return.

The probability of an event occurring is the frequency with which that event is expected to occur relative to all other events. A probability distribution sets out all the possible events with their expected frequencies of occurrence. Once the probability distribution is established, it is possible to apply a simple statistical technique in order to provide the measure of expected return.

Example 3.3: Measuring return

Following extensive analysis and forecasting, you have established that an investment in Mills Ltd offers the following probability distribution of returns, given different states of the economy:

State of the economy	Probability of state of economy	Return
Super boom	10%	50%
Boom	20%	30%
Normal	45%	10%
Recession	15%	-10%
Severe recession	10%	-20%

Note that the probabilities add up to 100%. Traditionally, probabilities are denoted by decimals between zero and one. All events are mutually exclusive. These conditions are all necessary in order to perform a probability analysis. The expected return is determined as in Table 3.5.

Table 3.5 Calculating the expected return (R_e)

State of the economy	Probability	Return	Expected Return
Super boom	10%	50%	5.0%
Boom	20%	30%	6.0%
Normal	45%	10%	4.5%
Recession	15%	-10%	-1.5%
Severe recession	10%	-20%	-2.0%
		R_e = expected return	12.0%

Page 3-11 The expected return on the investment is 12%. The formula is calculated as follows:

$$R_e = \sum_{j=1}^{n} P_j \times R_j \qquad \textit{(Formula 3.6)}$$

where: R_j = the j th possible outcome

P_j = the probability of the j th outcome

n = the number of possible outcomes

Measuring risk for a single share

Having earlier defined risk as the magnitude of expected difference between actual outcome and expected outcome, a commonly used statistic to measure this magnitude is the variance as calculated in Table 3.6, using the same data as Example 3.3.

Table 3.6 Calculating the variance

State of the economy	Probability	Deviation	Deviation squared	Variance
	1	2	3	4
	P_i	$R-R_e$	$(R-R_e)^2$	13
Super boom	10%	38%	14.44%	1.44%
Boom	20%	18%	3.24%	0.65%
Normal	45%	-2%	0.04%	0.02%
Recession	15%	-22%	4.84%	0.73%
Severe recession	10%	-32%	10.24%	1.02%
			σ^2 = Variance =	3.86%

Risk may also be expressed as the standard deviation rather than the variance. The standard deviation is simply the square root of the variance.

$$\sigma = \sqrt{\sigma^2} \qquad \textit{(Formula 3.7)}$$

where: σ = standard deviation

σ^2 = the variance

The standard deviation is $\sqrt{0.0386} = 19.65\%$

The formula for finding the standard deviation from the raw data may be written as follows:

$$\sigma = \sqrt{\sum_{j=1}^{n} P_j \times (R_j - R_e)^2} \qquad \textit{(Formula 3.8)}$$

where: P_j = the probability of the jth outcome

$(R_j - R_e)^2$ = the deviation squared of the jth outcome from the expected return calculated in Formula 3.6

The mean–variance rule

Once the expected return (measured by the weighted arithmetic mean) and the risk (measured by the variance or standard deviation) have been established, a general rule becomes apparent. It is commonly referred to as the *mean–variance* rule and is applicable to all investment decisions made by rational risk-averse investors. The rule holds that investment A will be preferred to investment B provided one of the following two conditions exist: Page 3-12

- either the mean expected return on investment A exceeds that of investment B and the variance of A is equal to or less than that of B,

- or the mean expected return on A exceeds or is equal to the expected return on B and the variance of B is greater than that of A.

It has given rise to the technique of *mean–variance analysis*, which is a widely-used way of making investment decisions. The rule is applicable to all risk-averse investors, regardless of the degree of their risk aversion, and is illustrated in Figure 3.4.

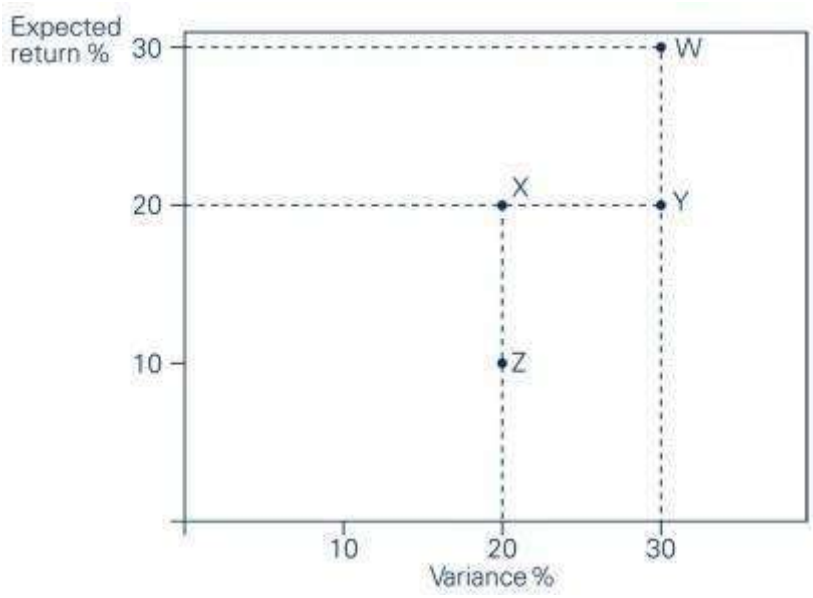


Figure 3.4 The mean–variance rule

The following can be seen from Figure 3.4:

- Investments W and X are always preferred to investments Y and Z respectively as they offer superior returns for the same risk in each case.
- Investment X is superior to investment Y because it offers the same expected return but is subject to less risk.

The mean–variance rule can now be applied to relevant investments. The measure of risk which will be used in this text will be the *standard deviation* (the square root of the variance) as it is more commonly used and is more easily applied to statistical manipulation arising from the normal distribution to be discussed in the next section.

3 INTERPRETING THE SUMMARY STATISTICS

The *expected return* is simply the weighted average of each return predicted for a given state of the economy. The risk associated with the expected return is measured

by the standard deviation of the return. The facts provided in Example 3.3 could be graphed as in Figure 3.5.

Figure 3.5 depicts a discrete probability distribution. Five mutually exclusive states of the economy have been identified. In fact, there are numerous possible states which could exist, each of that could be assigned a probability. The result could be graphed as a continuous probability distribution.

Properties of a normal distribution

Past company returns may follow a normal distribution. This means that in general they are symmetrically distributed around a central value, with decreasing frequency of occurrence as the distance above or below this central value increases. Page 3-13



Figure 3.5 Probability distribution of Mills Ltd rates of return

Using the data of Mills Ltd, a continuous probability distribution could be roughly drawn using the known characteristics of the normal distribution. From Figure 3.5 it is not absolutely apparent that the distribution has the characteristics of a normal or Gaussian distribution.

The normal distribution was first investigated in detail by the German mathematician Gauss. Its properties can be applied when results tend to be symmetrically distributed around a central value with increasing rarity of occurrence as the distance away from the central value increases. Where only five discrete probability states have been identified, the resulting histogram may not appear symmetrical. This is largely a result of the small number of discrete probability states and does not detract from acceptance of normal distribution characteristics as the basis

for determining the descriptive statistics. The most important properties of the normal distribution for an understanding and interpretation of risk as we have defined it are:

- Half the area under a normal curve is to the left of the expected return and half is to the right. This means that the curve always peaks at the expected return.
- Standard deviations demarcate points that cover the following area under the normal distribution curve:
  - Between + 1**σ** and -1**σ** = 68.3%
  - Between +2**σ** and -2**σ** = 95.5%
  - Between +3**σ** and -3**σ** = 99.7%

Using the standard deviation as the unit of measurement, it is possible to establish the area under the normal curve at any point by referring to Table E, at the end of the book. This will be explained later in the chapter.

These facts, together with the summary statistics obtained for Mills Ltd, are illustrated in Figure 3.6 from which the following observations can be made:

- The curve peaks at 12%, the expected rate of return. There is therefore a higher likelihood of a return of 12% than of any other return.
- There is a 68.3% chance that the return will be between –1σ and +1σ. Because the standard deviation has been calculated to be 19.65%, this means that there is a 68.3% chance that the expected return will be between –7.65% (12% – 19.65%) and 31.65% (12% + 19.65%). Conversely, there is a 31.7% chance that the return will be outside this range. As risk is the possibility that the outcome will be worse than expected, the chance of the return being below –7.65% is only half of 31.7%, that is 15.85%.

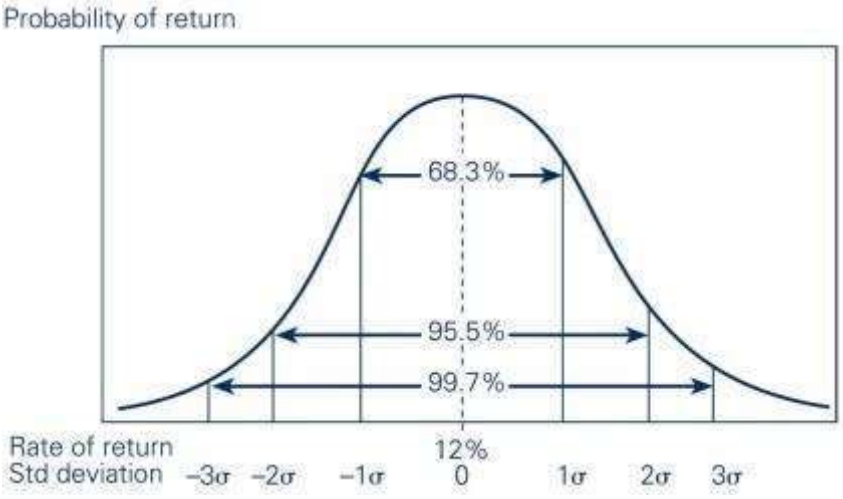


Figure 3.6 Continuous probability of Mills Ltd's rates of return: normal distribution

- There is a 95.5% chance that the actual return will fall between $+2\sigma$ and -2σ that is, between 51.3% (12% + 2 × 19.65%) and –27.3% (12% – 2 × 19.65%). Looking at the adverse side of the risk, we can say that there is a 2.25% chance, (half 100 – 95.5%), that the actual return will be below –27.3%. Similar numbers can be generated for the range -3σ to $+3\sigma$.

Comparison of single shares

We have focused on the summary statistics relating to a single investment opportunity. When faced with alternative investment opportunities, the investor who wishes to make a choice between two single investment opportunities needs some rationale on which to base the decision process. The strategy that should be adopted is illustrated in Examples 3.4 and 3.5 which follow.

Example 3.4: Choosing between two alternatives: identical expected returns

A financial analyst has gathered information regarding the performance of two listed shares over the last five years. Trinpak is listed under the paper and packaging sector, and Claycor under the building and construction sector.

The following summary statistics are generated:

	Trinpak	Claycor
Return to shareholders	22.0%	22.0%
Standard deviation of return	9.2%	15.4%

In this case, past performance may be used as a surrogate for expected future returns. Given that the expected return is identical for both investment alternatives, on what basis is a choice between the two alternatives made?

These two alternatives are represented in rough graphs in Figure 3.7.

It is immediately apparent that Trinpak has a much tighter distribution of returns. The probability of performing really badly, if past performance is used to predict the future, is much lower than that of Claycor. Faced with these two alternatives, it is likely that an investor would rather invest in Trinpak Ltd than Claycor Ltd. This is consistent with the mean–variance rule which implied that, in general, investors will select the alternative with lower risk when the expected returns are identical. Page 3-15

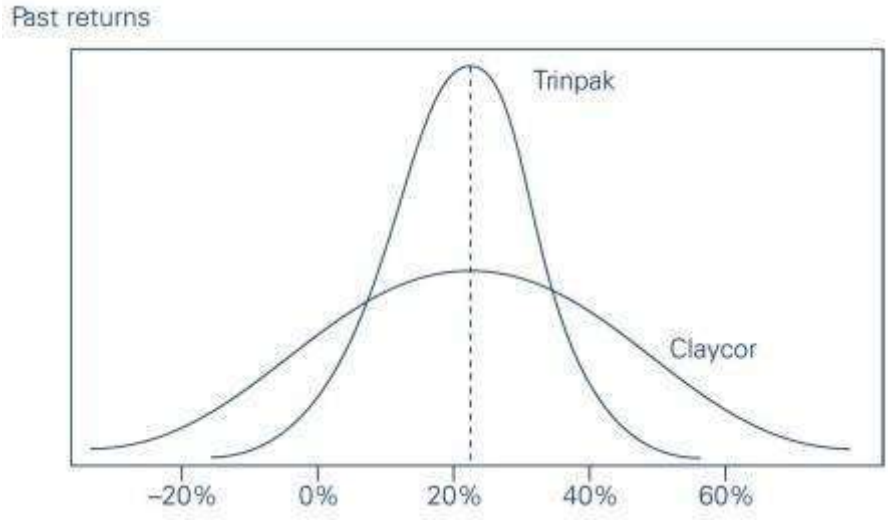


Figure 3.7 Distribution of returns: identical expected return

With investors showing a preference for Trinpak on the basis of the facts outlined, what is the likely result? The greater demand for Trinpak is likely to force its price upwards. Conversely, the lesser demand for Claycor would place downward pressure on its price until an equilibrium point, where return commensurate with the risk establishes the market price.

When comparing shares which have different expected returns, the decision is more complex. Looking at the above distributions an investor could surmise that Trinpak has the characteristics of a defensive stock e.g. a bank, insurance company, healthcare company or utility. These stocks are preferred during times of uncertainty and recession. Claycor, on the other hand, is not unlike a resource stock where the returns are poor in times of recession and high during a boom. A rush into resource stocks at the expense of more defensive stocks is a signal that an upturn in the world's economy is expected soon. The probability factors used in this chapter may have been skewed towards a more optimistic outcome.

Example 3.5: Selecting between two alternatives: different expected returns

An investor is considering the following two alternatives from which to select the most acceptable investment:

	Atlas Ltd	Brenco Ltd
Expected return	28%	20%
Standard deviation	14%	8%

Which investment should be selected?

This problem does not have a ready answer. The distributions are depicted in a rough graph in Figure 3.8.

It is apparent that Brenco Ltd offers a lower expected return, but this is not surprising as the degree of risk is lower than that of Atlas Ltd.

If an investor wishes to choose between these two shares, some basis of comparison is required. Using the properties of the normal distribution, two indicators, the co-efficient of variation and the z score, may be used.

Co-efficient of variation

The co-efficient of variation (CV) relates the units of return to the units of risk. It expresses the units of risk per 1% of return as follows:

$$CV = \frac{\sigma}{R_e} \qquad \textit{(Formula 3.9)}$$

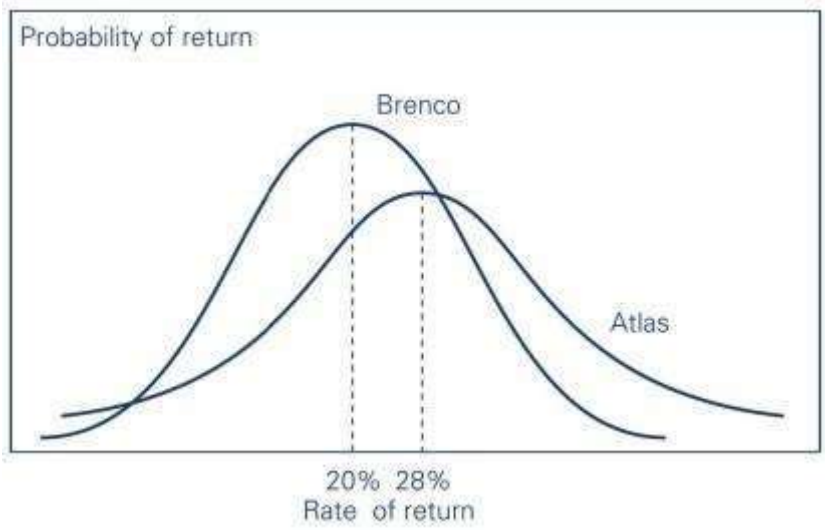


Figure 3.8 Distribution of returns: different expected returns

Using the CV, the following results are obtained for Atlas and Brenco:

Atlas	Brenco

CV	14/28 = 0.5	8/20 = 0.4

The results indicate that Atlas Ltd exposes the investor to 0.5 units of risk for each expected unit of return, while Brenco exposes the investor to only 0.4 units of risk for every unit of return. On a relative basis, therefore, Brenco seems to offer a better trade-off between expected return and risk. Note that this approach is not rigorous and serves only as an additional indicator to assist investors when selection between two alternatives with different expected returns is required.

The *z* score

The *z* score is used to help establish the probability of a return falling below a given level. For example, an investor may wish to determine the probability of the returns being zero or less. The following procedure would be adopted.

First, find the *z* score using the following formula:

$$z = \frac{x - \bar{x}}{\sigma} \qquad \textit{(Formula 3.10)}$$

where: x = the given rate being examined

$\bar{x}$ = the expected return (the mean)

σ = the standard deviation

Consult Table E – the standard normal distribution (at the end of the book) – and determine the area to the right of 0% by reading from the table. The figure obtained is the probability of returns falling between 0% and the mean. Then interpret the results.

Using the *z* score, the following results are obtained for Atlas and Brenco:

	Atlas	Brenco
Z score at 0%	$\frac{0-28}{14} = -2$	$\frac{0-20}{8} = -2.5$
Table reading	0.4772	0.4938
% Probability (0.5 – Table)	2.28%	0.62%

 Note that Table E is constructed for only one side of the symmetrical curve, as a z score table for the mirror image side would be identical. In order to establish the probability of returns being less than zero, the table reading is deducted

from 0.5 (the left-hand side of the distribution with which we are dealing). The probability is then expressed as a percentage. This is illustrated for Atlas Ltd in Figure 3.9.

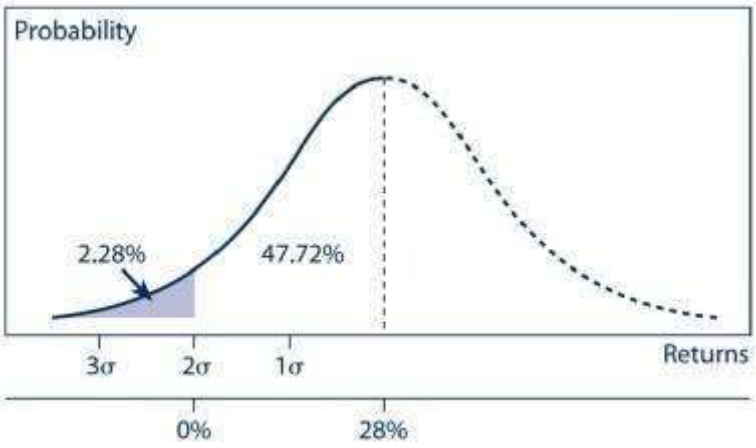


Figure 3.9 Using z scores

This approach once again does not provide perfect information, but allows investors to develop perceptions regarding the risk/return relationship.

It should be clear that 0% return is a critical point because once returns fall below this point the investor suffers not only a zero return on capital but also a loss of capital invested. Investors may, however, be sensitive to other points in the operation of probable returns. If, for example, an investor wishes to establish the probability of returns being below 15%, exactly the same procedures are followed.

Atlas has a lower probability than Brenco of generating returns below 15% and (as is apparent from the earlier calculation) it has a higher probability of returns below 0%. These indicators are consistent with expectations regarding risk and return.

	Atlas	Brenco
Z score at 15%	$\frac{15-28}{14} = -0.93$	$\frac{15-20}{8} = -0.63$
Table reading	0.3238	0.2357
% Probability (0.5 – Table)	17.62%	26.43%

We can use the Excel function =NORMSDIST to determine the standard normal cumulative distribution function rather than employ Table E. For example, if you enter =NORMSDIST(-2) in an Excel worksheet cell, the result will be 2.28%.

	A	B	C

32		Atlas Ltd	Brenco Ltd
33	Z score at 0%	-2.00	-2.50
34	% Probability that return < 0%	2.28%	0.62%
35	Excel function in Cell B34	=NORMSDIST(B33)	
36	Z score at 15% (rounded)	-0.930	-0.630
37	% Probability that return <15%	17.62%	26.43%
38	Excel function in Cell C37		=NORMSDIST(C36)

Page 3-18 We can use the =NORMSINV function in Excel to indicate the number of standard deviations from the mean which matches a given confidence level. For example, =NORMSINV(0.0228) will come up with an answer of –2 standard deviations.

Co-variance and correlation

Co-variance measures how share returns move together and specifies the strength of the relationship between share return movements. It indicates the extent that share returns co-vary. For example, we would expect that the share prices of resource companies would move together. In general, if the share price of Gold Fields goes up, then we would expect the share price of Harmony also to increase. The share prices of these companies tend to co-vary.

Mathematically, the co-variance between x and y is denoted as follows:

$$\text{Cov}(x,y) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) \qquad \textit{(Formula 3.11)}$$

where n is the number of observations.

If share returns tend to co-vary, then the actual returns will tend to be above or below their means for the same period and the product of this relationship will be positive. If share returns move in opposite directions, then the co-variance will be negative.

Cov(x,y) > 0, x and y will tend to move in the same direction
Cov(x,y) < 0, x and y will tend to move in opposite directions
Cov(x,y) = 0, x and y are independent. There is no relationship between the returns of x and y.

What does co-variance indicate? It is a number and it is difficult to interpret on its own. We use the **correlation co-efficient**, p_{xy}, which is a standardised version of the co-variance, to determine the relative strength of the linear relationship between x and

y. We need the covariance to determine the correlation co-efficient which is denoted as follows:

$$\rho_{xy} = \frac{\text{Cov}(x,y)}{\sigma_x \sigma_y} \qquad \textit{(Formula 3.12a)}$$

The correlation co-efficient is between -1 and +1. A correlation co-efficient of (+ 1) indicates that share returns are perfectly correlated. This means that a (+ 10%) share return by *x* is matched by a (+ 10%) return by *y*. If the share return of *x* is (–10%), then the share return of *y* will also be (–10%). A correlation of (–1) indicates that shares are perfectly negatively correlated. Therefore a (+ 10%) share return by *x* is matched by a (–10%) return by *y*. We can restate the above formula as follows:

$$\text{Cov}(x,y) = \rho_{xy} \sigma_x \sigma_y \qquad \textit{(Formula 3.12b)}$$

How do we calculate the co-variance? We will use the example set out in Table 3.7 and do it the long way before simply using the relevant Excel functions. We will come back to this example in Chapter 4 on Portfolio Management. Why is co-variance important from a risk perspective? If the returns of investments do not co-vary, it means that portfolio risk will be lower than if the returns do co-vary. We will expand more on this in Chapter 4.

In Table 3.7, we have two shares P and Q which offer the returns as set out in the table for each year. We calculate the mean return for P and Q and then set out the difference between the return and the mean for each observation, to comply with part of the above formula, i.e. $(x - \bar{x})(y - \bar{y})$. We then sum the product of the differences and calculate the average of Page 3-19 the product to determine the co-variance. We can use the Excel functions, COVAR (array 1, array 2) and CORREL (array 1, array 2) to determine the co-variance and the correlation co-efficient.

Table 3.7 Co-variance and correlation

	A	B	C	D	E	F	G	H
2		P			Q			PQ
3		a			b			a x b
4	Year	Return	Return-Mean		Return	Return-Mean		Product
5	1	40%	10%		32%	12%		1.20%
6	2	16%	-14%		8%	-12%		1.68%
7	3	44%	14%		30%	10%		1.40%
8	4	20%	-10%		14%	-6%		0.60%
9	5	44%	14%		30%	10%		1.40%
10	6	16%	-14%		6%	-14%		1.96%
11	7	42%	12%		28%	8%		0.96%
12	8	18%	-12%		12%	-8%		0.96%
13								
14	Sum	240%			160%			
15	Mean [sum/8]	30%			20%			
16			Covariance = AVERAGE(H5:H12)					**1.27%**
17			Covariance = COVAR(B5:B12,E5:E12)					**1.27%**
18			Correlation = CORREL(B5:B12,E5:E12)					**97.83%**

We have used the Excel functions to support the calculations. Note that we have determined the co-variance by using percentage returns. If using absolute numbers, the co-variance would be 127, but the correlation would remain at 97.83%. We will come back to this example in Chapter 4.

Value at Risk (VaR)

VaR answers a simple question. How much can a firm lose over a period of time (such as a day), under normal market conditions and subject to a given probability. A portfolio with a one-day 5% VaR of R50m means that there is a 5% probability that the portfolio will fall by more than R50m in any one day. Value at Risk is a useful risk management method used by banks, financial institutions, firms and investors to manage exposure to losses. Although volatility is useful to measure risk, it includes downside and upside movements as volatility indicates dispersion around a mean. Investors are not as concerned by dramatic upward movements in prices or returns although it is important to determine whether such movements are supported by improvements in operating performance. Although Warren Buffett's Berkshire Hathaway share price has been volatile, this was mainly due to dramatic upside movements in the share price. Investors are much more concerned about losses and about measuring a worst-case scenario. Investors wish to know the probability of losing money and VaR is a useful risk management tool that includes three aspects; a **time period** like a day, month or year, a **95% or 99% level of confidence** and a **specified investment loss** stated either in monetary terms or as a percentage. In practice, managers will often estimate 5% VaR.

Let's use an example and real financial data to explain VaR. An asset manager wishes to evaluate investing in emerging markets rather than investing in the USA but is concerned about downside risk and wishes to apply a 5% VaR. The average monthly return of investing in emerging markets from January 1988 to May 2023 was 0.742% per month and the standard deviation was 6.39%. Page 3-20

In terms of the normal distribution, the one-tail 95% VaR is equivalent to 1.6449 standard deviations from the mean[4]. If the distribution of emerging market returns is normal, then we could determine the VaR as follows:

$$\text{VaR} = \text{Mean} - (1.6449 \times \text{Standard Deviation}) = 0.742\% - 1.6449 \times 6.39\% = -9.769\%$$

Therefore, there is a 95% probability that monthly returns will be above –9.769% and a 5% probability that monthly returns will be equal to or worse than –9.769%. This means that the asset manager can expect a negative monthly return of *at least* –9.769% once every 20 months (1/20 = 5% of the time).

We can also determine VaR by ranking the historical data, assuming that the historical returns are representative of what could occur in the future. There are 425 months of returns between January 1988 and May 2023 and 5% of 425 is 21.25. If we rank the MSCI Emerging Markets' monthly returns from lowest to highest returns, then the lowest 22 monthly returns with associated months are as follows:

1	Aug-31-1998	-29.285%	12	Aug-31-1997	-12.841%
2	Oct-31-2008	-27.500%	13	Jan-31-2008	-12.594%
3	Sep-30-2008	-17.708%	14	Jun-30-1989	-12.248%
4	Oct-31-1997	-16.490%	15	Sep-30-2022	-11.902%
5	Sep-30-2001	-15.687%	16	May-31-2012	-11.668%
6	Mar-31-2020	-15.608%	17	Sep-30-2002	-10.967%
7	Sep-30-2011	-14.782%	18	May-31-2006	-10.764%
8	May-31-1998	-14.012%	19	Jan-31-1995	-10.707%
9	Mar-31-1990	-13.887%	20	Jun-30-1998	-10.694%
10	Aug-31-1990	-13.037%	21	Mar-31-2001	-10.282%
11	Sep-30-1990	-13.000%	22	Jun-30-1992	-10.252%

The 5th percentile is between -10.282% and -10.252% and we use Excel to determine this to be -10.273% [=PERCENTILE.EXC][5]. This is based on actual data and is a larger loss than the projected loss of -9.769% assuming a normal distribution. It is likely that market turmoil, disruptions and meltdowns are more common than indicated by a normal distribution. This is referred to as "fat tails" to describe the distribution and our results for emerging market returns support this assertion to some degree. In practice, there are more likely to be fat left tails implying a higher degree of

significant losses than predicted by the normal distribution[6]. VaR is a useful indicator of downside risk as in this case we can expect worse than a –10.273% return, 5% of the time which is about once in 20 months. The asset manager is now aware that he or she has a 5% chance of losing more than 10.273% in any single month. It is also true that –10.273% is the most optimistic loss and it does not capture a worst-case scenario. Page 3-21

Expected Shortfall

VaR reports the most optimistic measure of risk by reporting the smallest loss. It may be more realistic to report what the expected loss would be within the 5% range of worst possible returns. This is called the *expected shortfall* and we assume an equal probability for each negative return falling within the 5% most negative returns range. If we sum the lowest 21 returns set out above plus 0.25 of –10.252% and then divide by 21.25, we can determine an *expected shortfall* of –14.505%. This is a significantly greater loss than the negative 10.273% return disclosed with VaR. This is also called *conditional tail expectation* or *conditional value at risk*.

Figure 3.10 reflects all the monthly returns ranked from lowest to highest returns which is interesting as it indicates that there were 175 months in which emerging markets reported negative monthly returns from a total of 425 months. This is also a useful indicator of risk.



Figure 3.10 Emerging markets monthly returns ranked from lowest to highest: 1988–2023

Conclusions about VaR

Another way of determining VaR is to apply Monte Carlo simulation and we will explain the use of Monte Carlo simulation to evaluate the risk of capital projects in Chapter 10. We will come back to emerging markets later in the chapter. In this section, we have explained the use and purpose of VaR to indicate the probable monthly loss we can expect to experience 5% of the time. In order to manage downside risks, it is useful for an asset manager to know that there is a 5% chance that in any month the firm could experience a loss greater than –10.273% (or an expected shortfall of –14.505%) when investing in emerging markets. How does this compare to investing in a developed market like the USA? We will come back to this question later in the chapter.

Historical returns

When we determined an expected return in Example 3.3, we projected possible future returns and associated probabilities. When we analyse historical returns, we are simply given realised returns and time periods and we assume an equal likelihood for each period's return. This means that in order to determine an average return, we simply assume a *1*/*n* probability of each return. Therefore, we can determine an arithmetic average by dividing the sum of returns by the number of periods. If the actual returns for years 1 to 3 were –20%, Page 3-22 +40% and +10%, then the average annual return is 30%⁄3 = 10% and we implicitly assume that there is a 1/3 probability of each return occurring.

We can also use a compound annual growth rate[7] (geometric average) to explain market returns. When we report that a company's share price of R100 went up by 10% in the first year and then by another 30% in the following year, we normally apply a multiplicative factor so that at the end of Year 2, we would expect the share price to be R100×1.10×1.30 = R143. We will further assume in our example that the share price at the end of Year 3 is R133.10 so that the share prices and related returns are set out below:

Year	Price	Return	1+return
0	100.00		
1	110.00	10.000%	110.000%
2	143.00	30.000%	130.000%
3	133.10	-6.923%	93.077%
Sum of returns		33.077%	
Arithmetic average	1/3	11.026%	

In the case of share returns and other series that involve compounding, the geometric average (compound annual growth rate) may be more appropriate than an arithmetic average. Although the arithmetic average is 11.026%, the compound annual return is computed to be 10% [$(133.1/100)^{1/3} - 1 = 10\%$]. We can also set this out as the nth root of the product of 1+return for each year, as follows:

$$CAGR = \left((1 + R_1)(1 + R_2)\ldots(1 + R_n)\right)^{\frac{1}{n}} - 1 \qquad \textit{(Formula 3.13)}$$

$$(1.10 \times 1.30 \times 0.93077)^{1/3} - 1 = 10\%.$$

If we multiply our investment of R100 by a factor of 1.10 each year, we will achieve a value of R133.10 by the end of Year 3. In Chapter 2, we explained compounding and the formula of $FV = PV(1+r)^n$. We are essentially using this formula to determine the compound annual growth rate (geometric average) as follows:

$$\text{Ending value} = \text{Beginning value}(1+r)^n$$

$$(\text{Ending value}/\text{Beginning value})^{1/n} = 1+r, \text{ so } (133.10/100)^{1/3} = 1.10$$

What about dividends? Assume our company paid a dividend of R7 in Year 1 and a dividend of R10 in Year 2. No dividend was paid in Year 3. How do we compute returns? It is important to include the reinvestment of dividends in order to compare returns of companies and to compute the **total shareholder return (TSR)** that the company achieved per year. We assume that the dividend is paid at yearend and we use the dividend to purchase shares, so that the dividend of R10 in Year 2 will be used to purchase 0.06993007 (R10/R143) of one share for every share held at the end of Year 2. Prior to the dividend in Year 2, the investor has 1.06363636 shares, so the shares purchased with the dividend are 0.07438017 shares in Year 2 (1.06363636 x 0.06993007). If we start with one share at the end of Year 0, then the value at the end of Year 3 will be R151.47.

We have been able to purchase 0.06363636 shares in Year 1 and 0.07438017 shares in Year 2 with the dividends, which means we will have 1.13801653 shares at the end of Year 3. The shares, prices and values are set out as follows: Page 3-23

		0	1	2	3
Dividend per share			7.00	10.00	-
Shares purchased with dividend# [Dps/price x shares held prior to dividend]			0.06363636	0.07438017	-
Cumulative shares*	A	1.00000000	1.06363636	1.13801653	1.13801653
Price per share	B	100.00	110.00	143.00	133.10
Value	A×B	**100.00**	117.00	162.74	151.47

		0	1	2	3
End value/Beg value		1.5147		# Yr 1: 7/110 x 1 share = 0.06363636	
Compound annual return		14.84% [(151.47/100)$^{1/3}$ -1]		# Yr 2: 10/143 x 1.06363636 shares = 0.07438017	
* The firm owns 1.06363636 Sharesatthe end of Yr1 and owns 1.13801653 (1.06363636+0.07438017) Sharesatthe end of Yr2.					

Value at end of Year 3: 1.13801653 shares × R133.10 = R151.47

Compound annual return = $(151.47/100)^{1/3} - 1 = 14.84\%$

The arithmetic average returns based on changes in the prices and changes in the values of the shares (with the reinvestment of dividends) are as follows:

Arithmetic returns	0	1	2 3	Sum	Avs return
Price	10.00%	30.00%	-6.92%	33.08%	11.026%
Value (with dividend reinvestment)	17.00%	39.09%	-6.92%	49.17%	16.389%

The compound annual return is lower than the arithmetic average return. We have computed annual returns. However, we can compute the **holding period return** which represents the total return over a period of time. For example, the holding period return in our example is 51.47% over the three-year period.[8]

The compound annual return based only on price movements is 10% per year which assumes that dividends are not reinvested. If we include the reinvestment of dividends, then the compound annual return will increase and this incremental return due to the reinvestment of dividends is significant over long periods.

Let's go back to *arithmetic average returns* and *compound annual returns*. Which average is better? The compound annual return better describes the long-run performance of an investment. This is the actual return we earned from Year 0 to Year 3. If we wish to compare investments, then it is more relevant to use compound annual returns and this is what unit trusts, mutual funds and Warren Buffett use to set out relative investment returns. Arithmetic average returns are always higher than geometric returns due to volatility. Yet, the arithmetic mean can be useful as an unbiased predictor of next year's return. We have written more about the nature of returns in Appendix 3.1 whereby we demonstrate why compound returns are more reliable and accurate from an investment perspective than arithmetic average returns.

4 RISK AND RETURN IN FINANCIAL MARKETS

In this chapter, we have noted that investors will demand a higher return for taking on a higher level of risk. If a government bond is offering an expected return of 6% per year, then we would demand a higher return for investing in corporate bonds and an

even higher return for investing in ordinary shares. Why? A company's earnings are subject to greater economic risks and share prices can be volatile. The additional return that investors will demand for investing in ordinary shares rather than government bonds is called the equity premium or the market risk premium. We will come back to the equity/market risk premium in Chapter 7. In this section we would like to understand whether, historically, ordinary shares have offered higher returns than government bonds. The question of whether investors have been rewarded for taking on the additional risks of being invested in ordinary shares is important as the risk/return relationship is critical to the theory of finance.

In South Africa, investors in ordinary equities earned a real return of 7.0% per year over the period 1900 to 2022, a period of 123 years. A real return is what you earn after deducting inflation. Page 3-24

Investors in bonds earned a real return of 1.8% per year over the same period. Cash/Bills earned 0.9%. Over the period 1973 to 2022, South African equities earned a real return of 7.1% per year and bonds earned a real return of 2.4% per year. Cash/Bills earned 1.8%. You will note from Table 3.8 that South Africa offered the highest equity returns in the world over the period 1900 to 2022. Equities in the USA offered a return of 6.4% per year while Germany offered a return of 3.1%. Equity returns in Russia and China were deeply affected by war and revolution, which resulted in extensive financial losses and we have not included these countries in Table 3.8.

It is interesting to convert these returns into cumulative real returns for each country. One Rand (or equivalent) invested at the beginning of 1900 in ordinary equities would have grown to R4113.4 in real terms while R1 in bonds would have grown to only R9.0 in real terms.[9] This is a big difference and reflects the power of compounding. These accumulated sums tell the real story of investing over long periods of time. The extra 5.2% (7.0 – 1.8) return per year offered an investor an increase of R4 104.4 (4113.4 – 9.0) on the R1 investment compared to a bond investor who accumulated R9.0 only. Table 3.8 also discloses real returns for a shorter period from 1973–2022, which is 50 years.

Table 3.8 Real returns for equities, bonds and bills

Country	1900-2022 Equities Real Return	1900-2022 Bonds Real Return	1900-2022 Bills/Cash Real Return	Cumulative real returns Equities	Cumulative real returns Bonds	1973-2022 Equities Real Return	1973-2022 Bonds Real Return	1973-2022 Bills/Cash Real Return
	%	%	%	1900-2022	1900-2022	%	%	%
South Africa	7.0	1.8	0.9	4113.4	9.0	7.1	2.4	1.8
Australia	6.7	1.6	0.6	2912.2	7.0	5.5	2.7	1.9
Canada	5.7	1.9	1.3	914.6	10.1	4.8	3.5	1.7
France	3.4	0.0	-2.1	61.1	1.0	6.1	5.2	1.5
Germany	3.1	-1.6	-2.4	42.7	0.1	4.8	3.9	1.1
Italy	2.1	-1.1	-3.5	12.9	0.3	1.9	3.1	-0.6
Japan	4.2	-0.9	-1.8	157.7	0.3	2.4	3.3	0.2
Netherlands	5.0	1.3	0.3	403.9	4.9	6.7	3.4	0.8
Spain	3.4	1.6	0.1	61.1	7.0	2.9	2.7	0.6
Switzerland	4.5	2.0	0.7	224.5	11.4	5.3	2.5	0.3
United Kingdom	5.3	1.4	0.9	573.7	5.5	5.6	3.4	1.5
United States	6.4	1.7	0.4	2059.8	8.0	5.9	3.1	0.4
World ex-USA ($)	4.3	1.4	0.4	177.4	5.5	4.4	4.1	0.4

Source: Credit Suisse Global Investment Returns Yearbook, 2023 & Mike Staunton

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We would expect that equities should offer higher returns than bonds as equities involve higher levels of risk and bonds should offer higher returns than bills as bonds are subject to price volatility. The evidence in Table 3.8 is clear. Equities have outperformed bonds for each country and each period (except for Italy and Japan over the period 1973–2022) and bonds have outperformed bills for each country and for each period.

What about other periods? Table 3.9 presents the compound annual returns from investing in the All Share Index (ALSI), Bonds and Cash (Money Market Index) over the period 1987 to early 2010. Page 3-25

Table 3.9 Returns for equities, bonds and cash in South Africa

From 1987 to 2010	ALSI	All Bond	Cash
Compound annual return	16.13%	14.70%	13.16%
Standard deviation	24.43%	10.52%	3.89%
Sharpe ratio	14.52%	20.08%	14.73%

Source: Auret and Vivian (2010) "A comparative analysis of returns of various asset classes in SA for the period 1986-2010" African Finance Journal Conference, March 2010

The nominal return from investing in equities is higher than investing in bonds and cash – but not by much – resulting in a market risk premium of 1.43% that is much less than over the period 1900 to 2022. We will explain what the Sharpe ratio means later in the chapter.

What about risk? Do investors earn higher returns for undertaking higher levels of risk? Table 3.10 presents the returns from investing in equities and bonds and the associated volatilities as indicated by the standard deviations for each country and

asset class. Equities on average do earn higher returns than bonds and bills/cash but also report higher levels of volatility.

Table 3.10 Real average returns and associated standard deviations

Country	Equities Real Return	Equities Real Return	Bonds Real Return	Bonds Real Return	Equities Std Deviation	Bonds Std Deviation
	Geometric average	Arithmetic average	Geometric average	Arithmetic average		
1900 - 2022	%	%	%	%	%	%
South Africa	7.0	9.0	1.8	2.3	21.4	10.2
Australia	6.7	8.2	1.6	2.4	17.4	13.3
Canada	5.7	7.0	1.9	2.4	16.8	10.6
France	3.4	5.8	0.0	0.9	22.8	13.4
Germany	3.1	7.8	-1.6	1.5	31.1	15.5
Italy	2.1	5.9	-1.1	0.4	28.1	15.5
Japan	4.2	8.6	-0.9	1.5	28.9	19.2
Netherlands	5.0	7.0	1.3	1.9	21.1	10.4
Spain	3.4	5.5	1.6	2.4	21.5	13.0
Switzerland	4.5	6.3	2.0	2.4	19.3	9.6
United Kingdom	5.3	7.1	1.4	2.3	19.5	13.9
United States	6.4	8.3	1.7	2.2	19.9	10.7
World ex-USA ($)	4.3	6.0	1.8		18.8	

Source: Credit Suisse Global Investment Returns Yearbook, 2023 & Mike Staunton

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The geometric average returns for equities exceed the geometric average returns for bonds and the same is true for arithmetic average returns. The higher relative volatility of equity returns relative to bond returns is reflected in a much higher standard deviation for equities as compared to bonds. The standard deviation for South African equities is 21.4% whilst the standard deviation for bonds is 10.2%. Although not disclosed in Table 3.10, the standard deviation for South African bills is 5.9% as compared to an arithmetic return for bills of 1.0%. The measures of volatility support the premise that we would expect higher returns from equities in relation to bonds and higher returns from bonds in relation to Page 3-26 bills. Although there have been specific periods whereby bonds have outperformed equities, generally lower returns are aligned to lower volatilities.

Figure 3.11 determines the accumulated amounts in real terms from investing R1 in 1900 (or one year before 1900) in equities for selected countries. The annual compound returns in Table 3.8 do not adequately capture the differences in real accumulated values from investing in equities and reinvesting all income over 123 years.

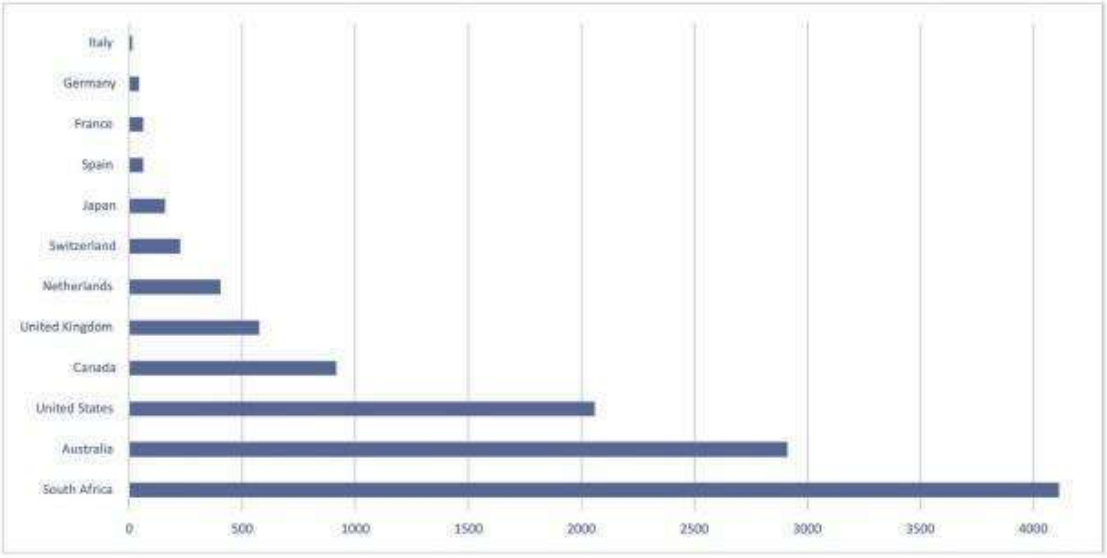


Figure 3.11 Cumulative amounts stated in real terms from investing 1 unit of local currency in equities (and reinvesting all income) from 1900 to 2022

In real terms, after inflation, investing R1 in 1900 would result in an accumulated amount of close to R4 113 by the end of 2022. If your great-grandfather had invested R1 000 in equities in 1900, and had bequeathed this to you, you would have received R4.113 million at the beginning of 2023. Yet, to be fair we should also include currency effects. Figure 3.11 depicts returns and accumulated amounts in each country's own currency.

When we take into account the fact that the Australian Dollar and the US Dollar have appreciated against the Rand, then Australian investors and USA investors have done better than South African investors. In fact, we often refer to Rand hedges, which are shares that should protect investors against any depreciation in the Rand and the JSE's market capitalisation is dominated by groups such as MTN, BHP Billiton, BAT, and Naspers that have significant international operations.

Figure 3.12 presents the cumulative returns of investing R1 in either the JSE ALSI, Bonds or Cash over the period between January 2004 and May 2023.

Although equities have been more risky, the returns to investors have been significantly greater than that of bonds and cash.

In another study by Professor Brian Kantor of Investec Securities Research, the mean return for the JSE ALSI (All Share Index) was 15.29% for the period January 1970 to July 2005 whilst bonds offered an annual return of 11.9%. The results are presented in Table 3.11. We have updated this to 2023. Investing in equities involved more risk but offered higher returns. However, although bonds implied higher levels of risk, they offered marginally better returns than an investment in cash over this

period. Investors in equities were also required to accept significant annualised falls in equity prices in some months. Accepting the upside was no problem (if you were invested) but it was necessary to live through the reversals in values as well. Page 3-27

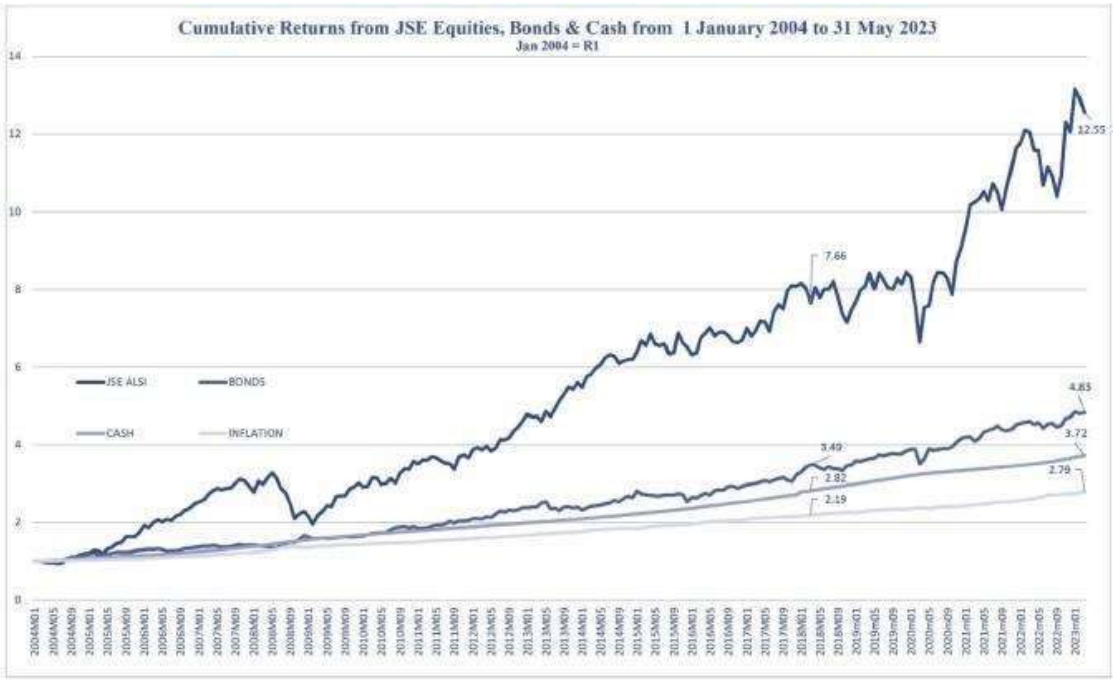


Figure 3.12 Investment of R1 in either equities, bonds or cash in Jan 2004, with reinvestment of income until May 2023

Source of data: Prof. Brian Kantor – Investec Securities

Table 3.11 Nominal annual returns and standard deviations

Jan 1970 to May 2023	JSE ALSI	Bonds (RSA)	Cash
[calculated monthly]			
Arithmetic average return (annualised)	17.48%	11.28%	10.04%
Standard deviation (annualised)	20.70%	8.23%	1.26%
Compound annual return (annualised)	15.31%	10.94%	10.06%
Maximum (monthly)	18.1%	12.0%	1.9%
Minimum (monthly)	-29.7%	-16.9%	0.3%
Jan 1970 to July 2005	**JSE ALSI**	**Bonds (RSA)**	**Cash**
Mean return (%)	15.29%	11.90%	10.54%
Standard deviation (%)	24.02%	12.89%	4.25%

Have equities always outperformed bonds? It depends on our starting date. In South Africa, bonds actually outperformed equities from 1990 to May 2005. That is a long time.

What about small companies? In the USA, since 1926, investing in “small value caps” offered investors an additional return of 3.6% per year. From January 1998 until December 2022, the MSCI Small Caps Index outperformed the MSCI World Index by 3.14% per year. From 1990 to 2010, the excess mean return for small caps in South Africa was found to be between 1.98% and 3.24%, relative to large market cap firms.[10]

Let’s compare the volatilities of different sectors. We will select the precious metals & mining sector and the food producers sector. Anglo American plc is a key component of the precious metals and mining index and Tiger Brands is a key component of the food producers sector. We have determined the monthly percentage changes in the mining and food producers indices. This is presented in Figure 3.13.

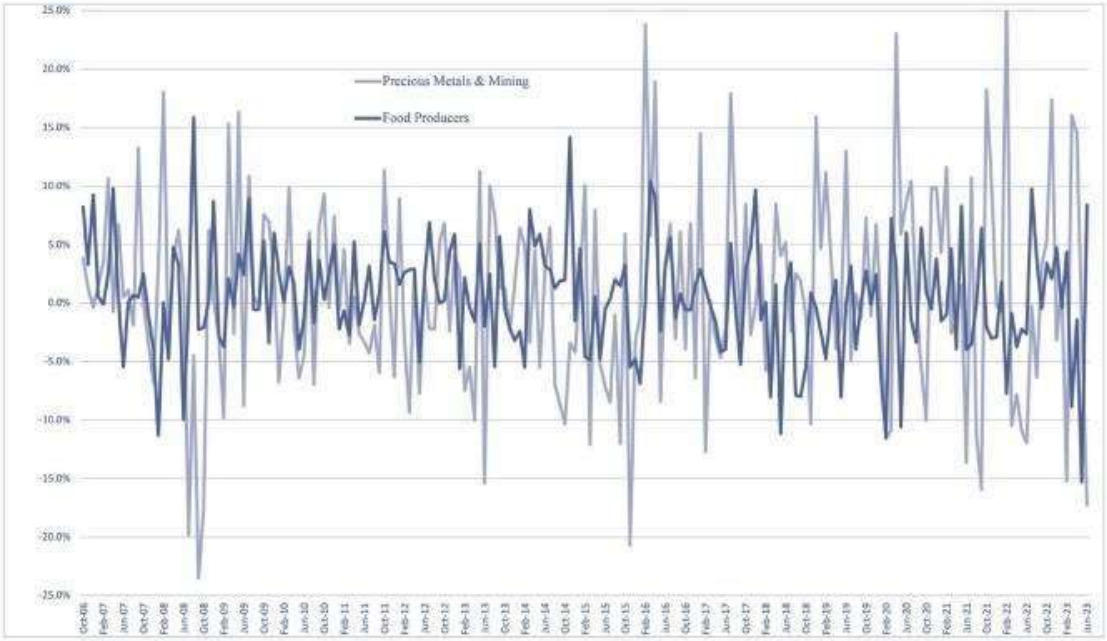


Figure 3.13 Monthly percentage changes in the Mining and Food Producers Indices: October 2006 – June 2023

We can clearly see that the Precious Metals & Mining Index is more volatile than the Food Producers Index over the period. Therefore, if we are investing in companies that form part of the Precious Metals and Mining Index, we need to be prepared to accept the additional volatility and risk that comes with investing in these companies. The standard deviation of the Precious Metals and Mining Index was 8.5% per month whilst the standard deviation for the Food Producers Index was only 4.7%.

Let’s evaluate the share price movement and volatility of Anglo American plc which is a company that forms part of the Industrial Metals & Mining Sector on the

JSE. The group is one of the world's leading resource companies and Figure 3.14 depicts the movement in Anglo American's share price from June 2006 to June 2023.

The movement in share price initially reflected the increase in commodity prices that was driven by Chinese demand. Then the economic crisis struck in the latter half of 2008 and the share price fell sharply from a high of R545.45 in June 2008 to R140.62 within a period of eight months. Although recovering in 2011, the price of Anglo American collapsed and hit R62.71 in January 2016 due to concerns about significantly lower growth in China, and the potential fall in demand for resources going forward.

The dramatic fall in Anglo American's share price indicated the risks of investing in equities. Although equity markets have recovered, we can see that the share price in June 2023 is only back to levels applicable in June 2008. Page 3-29



Figure 3.14 Movement in the Anglo American share price: June 2006 – June 2023

When we plot the share price, we do not obtain a clear picture of the underlying volatility in Anglo American's share price. In Figure 3.15, we plot the monthly percentage changes in Anglo American's share price over the same period.



Figure 3.15 Monthly percentage changes in the Anglo American share price: July 2006 – June 2023

Page 3-30 The graph of changes in the share price indicates the increased volatility around the time of the economic crisis. Greater volatility in the share price occurred during 2015 to 2017 due to lower economic growth in China, market turmoil and a significant fall in the oil price and then subsequently by the strong recovery in the global economic outlook. Over the period July 2006 to June 2023, the standard deviation was 11.106% per month. The average monthly return was 0.873%. This means that 68.3% of the monthly percentage changes in the share price were in the range of –10.233% to +11.979% (0.873% – or + 11.106%). If you invest in equities then you do need to accept the associated volatility in returns. However, the evidence indicates that investors (over the long term) do earn higher returns for accepting the higher risks of investing in equities.

Emerging markets

The growth in emerging markets has been phenomenal. Future growth will mainly occur in emerging markets such as China, India, Indonesia, Vietnam, Mexico and Brazil. Emerging markets is also where 70% of the world's population lives and South Africa is an emerging market. Emerging markets produced an annualised return of 10% per year in the decade to 2010 whilst for the developed world; this was close to zero. However, in the three years to October 2014, emerging markets reported lower returns than developed markets. Yet, despite this, Dimson, Marsh and Staunton of

London Business School write in the 2014 *Credit Suisse* Global Investment Returns Yearbook that "from 2000 to 2013, the terminal wealth accruing from investing in emerging markets was almost twice that from an equivalent investment in developed markets." From early 2016 until February 2018, emerging markets staged a significant rally in relation to developed markets, which also performed strongly. Yet from 2018 until 2023, emerging markets performed poorly in relation to developed markets. Figure 3.16 sets out values based on the MSCI indices for emerging markets and developed markets (MSCI World Index) since 1987. An investment of $100 in 1987 would have grown to $958.53 by 31 May 2023, whilst the same investment of $100 in developed markets would have grown to $686.43.

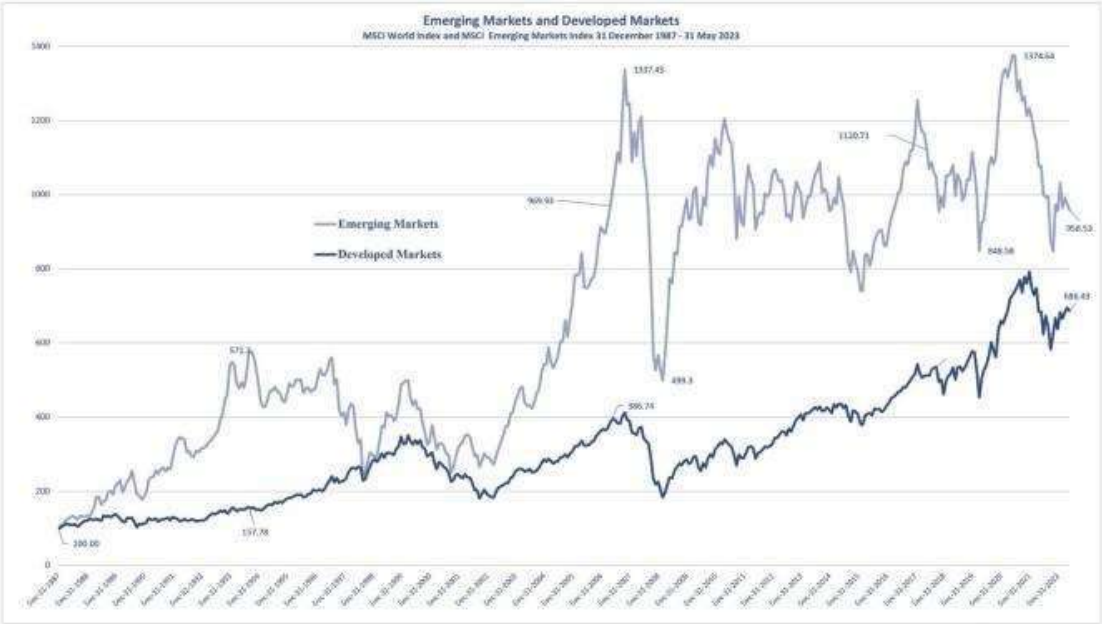


Figure 3.16 Emerging markets and developed markets: performance from 31 December 1987 to 31 May 2023

Page 3-31 What about risk? Emerging markets have always been seen as more risky. Is this true? The emerging markets index has generally been more volatile and emerging markets reported an annualised standard deviation of 22.13% over the 35 years to 2023 as compared to developed markets which reported an annualised standard deviation of 15.07%. The beta of emerging markets to developed markets is 1.30.[11] Investors should expect higher returns from investing in emerging markets as there is more risk there.

Table 3.12 presents summary statistics for emerging markets and developed markets. The higher relative arithmetic average and compound average returns reflect the higher risks of investing in emerging markets. We have included a Value at Risk

measure (by applying the =NORMDIST Excel function) and the data indicates that there is a 4.63% chance of losing more than 10% in any one month by investing in emerging markets but there is only a 0.77% chance of losing more than 10% in any one month by investing in developed markets.

Table 3.12 Summary statistics for emerging markets and developed market returns Jan 1988 to May 2023

	B	C	D	E	F	G
477	**Summary statistics**				**Emerging Markets**	**Developed Markets**
478	Arithmetic average Monthly return				0.742%	0.550%
479	Standard Deviation				6.39%	4.35%
480	Compound average return				0.533%	0.454%
481	Annualised compound average return				6.59%	5.59%
482	Annualised Standard Deviation				22.13%	15.07%
483	Probability of losing 10% or more in any month#			=NORMDIST(-0.1,F478,F479, 1)	4.63%	0.77%
484	Probability of losing 10% or more in any month based on historical data				5.41%	1.65%
485	Lowest 10% of returns in any month start at this return*			=PERCENTILE.EXC(F38:F462,0.1)	-7.28%	-4.92%
486	Expected shortfall		[average return within the bottom 10% of returns]		-11.50%	-8.14%
487	#Assumes future returns will follow a normal distribution			*Based on actual data and assumes future returns will be similar to past returns.		

From a diversification perspective, investors can reduce the risk of their portfolio by investing in both developed markets and emerging markets. We will come back to the issue of diversification and risk in the next chapter.

Volatility and time periods: the Rip van Winkle solution to risk

Volatility is often time dependent. The shorter the time interval, the greater the volatility will be on an annualised basis. Annual holding periods generate greater volatilities than 5-year or 20-year holding periods. A study by Firer and Staunton (Investment Analyst Journal, 56, 2002) found that an annual holding period resulted in a standard deviation of 23% for equities. However, as we increase the investment period and evaluate a 5-year holding period, we find that there is a significant fall in the standard deviation to 9.9%. If we follow the Rip van Winkle solution to risk and fall asleep for 20 years, then we will find that the standard deviation for a 20-year holding period is only 5.5%. If you invest for 20-year holding periods, then equities have always out-performed bonds. The markets have offered maximum positive real returns of close to 100% and negative real returns of close to –60% over a one year holding period. Over a 5-year holding period, the maximum positive and negative returns are 20% and –20%. Over a 20-year holding period, the maximum real return is about 12% and the maximum negative return is close to zero. If we go back to Figure 3.14, if you had invested in Anglo American in June 2008, went to sleep

and you only woke up in June 2023, you would have looked at the share price and it would have hardly moved. You would have missed a roller coaster ride of volatilities and price movements.

What does Warren Buffett think?

According to Buffett, volatility is not a valid measure of risk and an investor should try to rather understand the business and the volatility inherent in the business. The risk that counts is not beta but the possibility of permanent loss. The quality of the business determines the level of risk. The competitive strength of Coca Cola and Apple reduces the risk of investing in these companies relative to other companies that do not have a sustainable competitive advantage but may have similar betas.

What do we think? We think that the importance of volatility may be overstated and we agree that a more relevant measure of risk relates to the probability of failure. We will come back to this when we evaluate project risk in Chapter 10 and we focus not only on measures of volatility but also on the strategic context of an investment. Yet our view is also more nuanced. Why? We consider that volatility may be relevant for determining the probability of business failure. It depends on the company.

Firstly, let's assume a company is mining and producing platinum and the company has high levels of debt. Platinum prices have been volatile and therefore the company's share price has also been volatile. Even though the market is expected to recover, the banks may force the winding up of the company before markets are able to recover. If the share price is undervalued at that point, then the company may not be able to raise equity finance to prevent bankruptcy. Remember that the company is expected to recover but due to price volatility and fixed borrowing costs and fixed operating costs, the company is incurring losses. A company with little debt is far less likely to fail as it does not have to meet fixed borrowing costs or arrange debt refinancing in volatile times. Corporate survival is the key factor as this enables a company to be around to take advantage of investment opportunities when the market does recover. The question we should ask is whether volatility endangers the existence of a company. If not, then volatility may not be that important.

Investors like Warren Buffett in fact welcome volatility as it offers opportunities to purchase good businesses at distressed prices. This does not mean that there is no risk – at least for investors who are forced to sell in a time of volatility due to pressure, placed by the lenders. However, we also agree that there is much more to risk than simply evaluating volatility, particularly volatility over short time intervals. Over

longer time intervals, volatility falls but the risk of business disruption rises. We will come back to the issue of volatility and risk in Chapter 10. We need to understand that Buffett does not like debt financing. Yet betas are affected materially by the level of debt financing. A highly geared company will tend to have high betas. We will analyse the impact of debt on betas in Chapter 7. Buffett also does not believe necessarily in diversification although in this case his company, Berkshire Hathaway, does hold a diversified portfolio. The expression, "Don't put all your eggs in one basket' is replaced by Mark Twain's alternative advice: 'Place all your eggs in one basket but watch that basket".

Risk-adjusted measures of performance

In Table 3.10, equities offer higher returns but also higher volatility than bonds and bills. However, the question is whether returns are enough to offset the additional risks of equities. Further, it is useful to compare portfolios and unit trusts on the basis of risk-adjusted returns. We have already explained the co-efficient of variation which is equal to an investment's standard deviation divided by its mean return. We will now explain a few other risk adjusted measures often used by fund managers to evaluate performance relative to risk. Page 3-33

All that glitters is not gold…. and a few other thoughts.

What about gold? Whilst gold can be a hedge, on its own it has been a terrible investment over the longer term. From 1900 until 2022, gold produced a real return of only 0.76% per year and an investment of $1.00 in gold in 1900 would have reached a value of $2.50 by 2022 as compared to $1 in equities which would have reached a value of $2024.[12] Yet the volatility (standard deviation) of gold was 17% per year. No wonder Warren Buffett does not like gold.

What about commodities such as wheat, cotton, coffee, copper, aluminium etc? The average real return of 29 commodities is actually slightly negative over the period 1900 to 2022. If commodity investments are equally weighted and rebalanced each year, then the real return becomes 2.04% per year at a standard deviation of 12.47% due to the diversification effect. Furthermore, we did not consider storage, insurance, or other carrying costs. Commodity futures that are equally weighted are a little more promising and offered a return of 3.28% per year from 1877 to 2022 but with a standard deviation of 17.55%. Commodities can be a hedge against equities and inflation.[13]

It is important to remember that only a small group of companies drive market performance. A study by JP Morgan[14] found that 2 out of 3 US companies underperformed the Russell 3000 index over the period 1980 to 2014. Forty percent of all listed companies produced negative results. The percentage of extreme winners is only 7%. This makes selecting a concentrated portfolio extremely risky, but potentially very profitable. However, one will be sure that investing in a market index will capture at least part of these returns.

Treynor ratio

This ratio measures the excess return per unit of systematic (unavoidable) risk as indicated by a portfolio's beta.

Treynor ratio = (average return of portfolio – average risk-free rate)/Beta of portfolio

The Treynor ratio is particularly useful to compare unit trusts that may invest in similar sectors. A fund with a higher Treynor ratio has achieved a higher return adjusted for market risk.

Sharpe ratio (reward to variability ratio)

Whilst the Treynor ratio measures excess return per unit of systematic risk, the Sharpe ratio measures excess return per unit of total risk as indicated by an investment's standard deviation.

Sharpe ratio = (average return of asset – average risk-free rate)/Standard Deviation

Auret and Vivian (see Table 3.9) conclude, on the basis of the Sharpe ratio, that bonds offered a superior risk-adjusted return than equities over the period 1987 to 2010.

Sortino ratio

This is similar to the Sharpe ratio but this ratio only evaluates downside risk.

Sortino ratio = (average return of asset – minimum acceptable return)/downside deviation

Page 3-34 Downside deviation is the standard deviation of negative asset returns. This measure assumes that investors are not as concerned with upside volatility and are mostly concerned with downside risk. Rather than use the average risk-free rate, we can set a target return or minimum acceptable return.

Jensen's Alpha

Portfolio managers are often evaluated on their ability to generate positive Alpha. The Capital Asset Pricing Model (CAPM), which we will cover in Chapter 4, is used to determine a theoretical risk-adjusted required return.

$$\text{Alpha} = \text{portfolio return} - (\text{risk-free rate} + \text{portfolio beta} \times (\text{market return} - \text{risk-free rate}))$$

If a portfolio generates a return that is higher than its risk-adjusted required return, then the portfolio manager will have generated positive Alpha. Essentially, the manager has generated an excess return in relation to the risk-adjusted required return.

FROM THE REAL WORLD – UNIT TRUST FUNDS

The main objective of **Coronation Fund Managers** in introducing its **Global Absolute Strategy** fund on 1 August 1999 was to achieve low risk and real returns. Since inception, Coronation has achieved an average annual return of 14.0% per year up to 31 March 2023 with an annualised standard deviation of only 8.0%. The fund's latest 10-year return was 8.9% per year. Its lowest monthly return was –8.1% and the maximum drawdown was –11.5%. The Sortino ratio was 1.7. The fund was able to protect investors from significant drawdowns and market falls experienced during the period June 2002 to April 2003 and during the global financial crisis period of November 2007 to February 2009 as well as the period April 2022 to September 2022. Coronation states in its fact sheet:

> The Coronation Global Absolute Strategy targets positive real returns with an overriding focus on limiting downside returns or portfolio losses. Therefore, capital preservation in real terms is equally important to return optimisation.

In order to achieve its investment objectives, the fund is able to invest in South African and foreign equities, South African and foreign fixed interest securities, cash, South African and foreign property, commodities and hedge funds.

The **Allan Gray Balanced Fund** on 30 June 2023 had a fund size of R175.0 billion. The fund factsheet states:

> The Fund invests in a mix of shares, bonds, property, commodities and cash. The Fund can invest a maximum of 45% offshore. The maximum net equity exposure of the Fund is 75% and we may use exchange- traded derivative contracts on stock market indices to reduce net equity exposure from time to time. Returns are likely to be less volatile than those of an equity-only fund.

Since inception on 1 October 1999, the fund has earned an annualised return of 15.0% per year and its latest 10-year annualised return is 9.2% per year. The annualised monthly standard deviation is only 9.4%. The maximum drawdown was 25.4%.

A low-cost way of getting access to the JSE is to invest in Index Funds such as the **Satrix ALSI Fund** which is offered by the JSE and simply tracks the performance of the FTSE/JSE All Share Index, but at a low cost. The annual return over the five years to 31 July 2023 was 9.65% per year. The Allan Gray Equity Fund reported a return of 7.6% per year (after fees) over the five years to 31 July 2023 and the Coronation Equity Fund reported a return of 8.2% (after fees) for the five years to 31 May 2023. The Coronation and Allan Gray Equity Funds

include direct exposure to foreign equities which is not so for the SATRIX ALSI fund. Page 3-35

What returns did Warren Buffett achieve with Berkshire Hathaway?

Warren Buffett has been called the greatest investor in history. An investment of $100 in his company, Berkshire Hathaway, in 1964, would have grown to $3 782 808 by the beginning of 2023. An investment of $100 in the S&P500 index (which reflects the largest 500 listed firms in the USA) would have grown to $24 380 over the same period. What return did Buffett achieve relative to the S&P500 over the period?

The compound annual return of Berkshire Hathaway was 19.93% from 1965 to 2022 as compared to 9.94% for the S&P500. This represents a difference of almost 10% per year over 59 years. Yet, this does not tell the full story. There are dramatic kinks in performance over the years. This is set out in Table 3.13.

Table 3.13 Berkshire Hathaway and S&P500 returns

Berkshire Hathaway (Warren Buffett)		BH (Buffett)	S&P500	Difference
1965 - 2022	Compound annual gain	19.93%	9.94%	9.99%
Cumulative sum at end of 2022 from investing $100 at end of 1964		3,782,808	24,380	3,758,429
1965 - 1991	Compound annual gain	27.67%	10.36%	17.31%
1992 - 2002	Compound annual gain	20.86%	9.19%	11.67%
1992 - 2022	Compound annual gain	14.37%	9.62%	4.75%
2002 - 2022	Compound annual gain	9.06%	8.02%	1.04%

Note: These results are based on the annual returns set out in the Berkshire Hathaway(BH) 2022 annual report. BH presents the compounded annual returns as 19.8% and 9.9% for BH and S&P500 respectively for the period 1965-2022. The cumulative amounts are reported to be $3787464 and $24708 which differ very slightly to our results probably due to the rounding of returns.

Note: These results are based on the annual returns set out in the Berkshire Hathaway(BH) 2022 annual report. BH presents the compounded annual returns as 19.8% and 9.9% for BH and S&P500 respectively for the period 1965-2022. The cumulative amounts are reported to be $3787464 and $24708 which differ very slightly to our results probably due to the rounding of returns.

In the period 1965 to 1991, Buffett's compound annual return was 27.67% per year as compared to 10.36% for the S&P500, an outperformance of 17.31% per year! However, for the period 1992 to 2022, his compound annual return was 4.75% per year higher than the S&P500. If we just focus on the period from 2002 to 2022, we see that he outperformed the S&P500 by only 1.04% per year. Why is this? As Buffett has accumulated capital, it has become increasingly hard to find investments that outperform the S&P500 by a significant amount. Buffett stated that "a fat wallet is the

enemy of superior performance" and Berkshire has significant funds under management.

Figure 3.17 sets out the annual returns of Berkshire Hathaway and the S&P500. As we can see, Buffett's outperformance occurred mainly in the years 1966 to 1994.

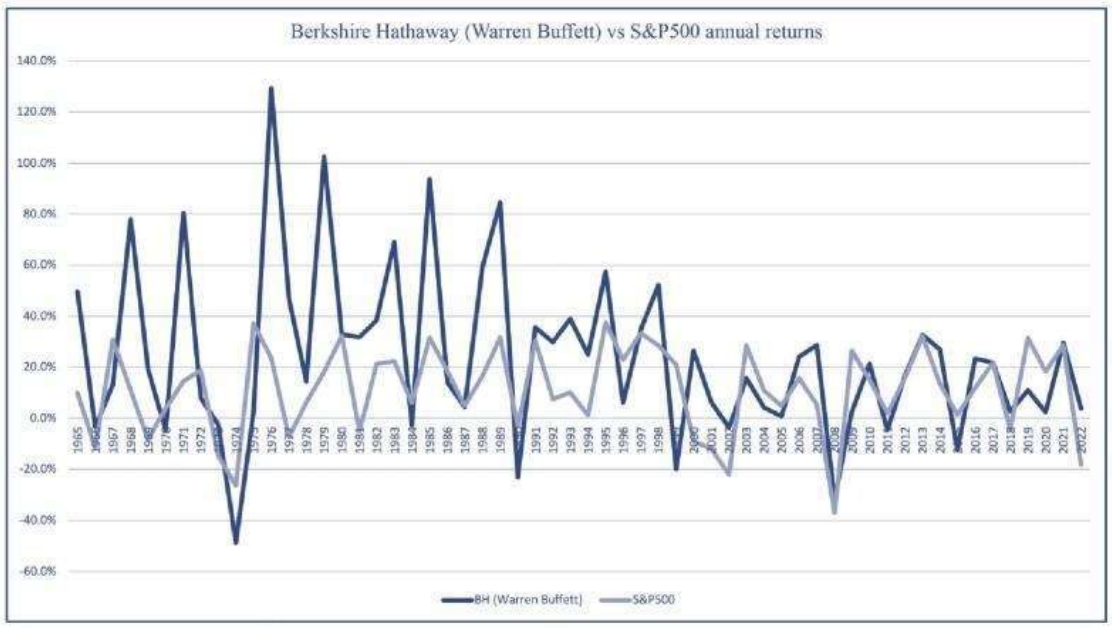


Figure 3.17 Berkshire Hathaway and S&P500 annual returns: 1965–2022

Page 3-36 Berkshire Hathaway has also indicated higher volatility of returns but this volatility relates mainly to dramatic upside movements in seven years over this period. This indicates one of the limitations of employing volatility to measure risk. Downside movements were relatively limited in relation to the S&P500 except for 1974. After 2002, performance was more closely aligned to the S&P500, as was volatility. Berkshire Hathaway is a conglomerate with interests in insurance, energy/utilities, railroads, industrial manufacturing, consumer products, banking, and investments in companies such as Apple, Coca Cola, Bank of America, HP, Occidental, Citigroup and Paramount. Will Buffett be able to offer returns significantly above the S&P500 in the future? We think that outperformance of more than 2–3% is unlikely. But over a long period of time, even one percent difference can make a large difference in the value of a portfolio. The price of Berkshire Hathaway (BRK-A) on 28 August 2023 was $539 605 per share!

SUMMARY

All investment decisions need to consider the required return, the expected return, and the estimated risk. The risk to which a company is exposed can be classified into two

distinct categories. Firstly, there is *business risk*, measured by the degree of operating leverage. Secondly, there is *financial risk* for a firm which has introduced fixed-interest debt into its capital structure. Financial risk is measured by the degree of financial leverage. The total risk is a product of these two categories and is expressed by the degree of combined leverage.

For the investor, a statistical technique is used for expressing the risk of an investment as the variation of expected returns, quantified in the form of the *standard deviation of expected returns*. As a result, two fundamental parameters in financial management, namely the expected return and the risk of an investment, can be captured in two summary statistics, the *arithmetic mean* and the *standard deviation*. Yet, from an investment perspective, the determination of a compound return (geometric mean) may be more relevant.

These summary statistics may be used to gain greater insight into the probabilities of expected outcomes. In order to make such interpretations, a normal distribution of expected returns is assumed and the properties of such distributions are used for further analysis. The *co-effcient of variation* and the *z scores* enable comparison to be made between two individual shares with differing expected returns and risk profiles.

An analysis of equity returns and bond returns indicates that investors do earn higher returns for taking on the higher risks of investing in equities. Risk adjusted measures of performance can be used to compare and rank the risk adjusted performance of funds.

The foundation has now been laid for further consideration of risk and return in the context of a portfolio of shares. This chapter has simplified matters in assuming that the risk and return information about individual shares is relevant for deciding between individual shares. The next chapter provides the rationale for holding a portfolio of shares and further develops the concepts discussed in this chapter.

APPENDIX 3.1 A BEHAVIOURAL FINANCE PERSPECTIVE ON RISK AND RETURN

The big money is not in the buying and the selling.... but in the waiting. Charlie Munger. Buy right and hold tight. John Bogle. No wise pilot, no matter how great his talent and experience, fails to use a checklist. Charlie Munger. The most important quality for an investor is temperament, not intellect. Warren Buffett.

In this chapter we have analysed market risk and return and we have demonstrated that although equities involved higher risks, over time equity markets have offered significantly

higher returns. What about individual investors? What were their returns like? In this section we wish to analyse Page 3-37 how behavioural finance[15] may explain investing behaviour. We reveal that our human traits and biases may increase the risk of investing in equities beyond what is indicated by market volatility. It seems that the risk of markets is compounded by the frailties of being human.

The evidence is that individual investors underperform a market index such as the ALSI or the S&P500. Barber and Odean (2000)[16] found that the average individual investor lost 3.7% per year on a risk-adjusted basis. This finding is supported by the results from later studies. The financial markets research firm, DALPERS, reported underperformance by investors of 2.84% per year relative to the S&P500. An investment in 1993 of $100 000 would result in the average equity fund investor accumulating $721 701 by 2022. Meanwhile, an undisturbed investment of $100 000 in the S&P500 in 1993 would have reached $1 585 839 by 2022.[17] In another study, Morningstar found that investors earned 1.67% per year less than their funds over a period of 10 years to 2022. The funds achieved a return of 7.71% per year whilst the investors in these funds earned a return of 6.04% per year. The performance of individual investors is important because increasingly in terms of defined contribution retirement plans, investors are responsible for the investment decisions of their retirement funds. Why do individual investors underperform the market index? The main reasons are:

- Overtrading of shares resulting in high transaction costs
- Selling winning shares and holding onto losing shares
- A high concentration in a few shares resulting in undiversified portfolios
- Investing in the investor's employer's shares, and local companies rather than investing internationally in order to achieve diversification
- Investing in shares on the basis of recent past performance
- Buying into bull markets and selling into bear markets and trying to time the market
- Over optimism that a rising trend in prices will continue or alternatively displaying a high level of pessimism when market prices are falling
- Investors' buying or selling decisions are often overly influenced by favourable or unfavourable media attention
- An inability to differentiate a great company from a great investment. This is called representativeness bias. The current share price may already be reflecting the fact that a company has an outstanding management team, great products and is reporting excellent financial results

- The application of mental accounting to funds may lead to irrational decisions in respect to risk and return decisions as well as portfolio construction
- Overconfidence that the investor is able to accurately forecast future prices, cash flows, dividends and/or earnings of companies
- The conservatism bias may lead investors to under-react to changes in earnings and prospects and then later over-react to further revisions
- Herding and following the crowd is fraught with danger particularly during asset bubbles. Sir Isaac Newton, one of the most brilliant minds in history, lost his wealth in the collapse of the South Sea Bubble and wrote: *I can calculate the motion of heavenly bodies, but not the madness of people.* Warren Buffett writes that one should be greedy when others are fearful and be fearful when others are greedy.

Page 3-38 There is another issue that is not captured above. Individuals tend to be underinvested in equities and over the long term this will place their portfolios and perhaps even their retirement at risk. In terms of **prospect theory**, investors are more sensitive to losses than gains. Investors will require a 50% chance of a gain of about R2 500 in order to offset a 50% chance of a R1 000 loss. This is also called loss aversion which implies that investors are averse to locking in losses. This may imply that the required premium for investing in equities above bonds or cash may be too high. Individuals may be reluctant to invest in equities as this will probably mean incurring short-term losses if investors monitor investments continuously. Fund managers often focus on "capital preservation" as a key component of their investment fund's investing strategy as this aligns with what individual investors value.

Studies indicate that following and buying into the trend in prices may offer short-term returns for individual investors but in the longer term, there is mean-reversion as share prices on average tend to reverse after four years.

In terms of the disposition effect, investors are more likely to sell shares that have gone up in price ("winners") and are more likely *not* to sell shares that have fallen below cost, ("losers"). If an investor buys a share for R100 and it falls to R70, then the investor is likely to hold on hoping that the share price will recover so that he or she can at least break-even. If the share rises from R100 to R130, then the investor is more likely to sell in order to lock in a gain. These actions may not be rational and evidence indicates that investors that hold losers and sell winners will lose on average. There is a saying that one should "cut your losses and let your winners run" at least for a few years. It is also not tax efficient as the investor owes capital gains tax on any realised gain.

Investors may be overconfident about their abilities to select winning shares. This overconfidence bias may lead to a concentrated portfolio which increases risks due to a lack

of diversification. Overconfidence in the ability to pick shares means that an investor may determine that diversification is not necessary. It may also mean that the investor becomes emotionally attached to particular companies which makes it hard to sell a particular share. Overconfidence may lead to too much trading and lower returns. Investors may invest in shares that are most familiar (in terms of the familiarity bias and home bias) and this may limit the performance of their portfolios and result in undiversified portfolios. Applying a representativeness heuristic (shortcut) may mean, for example, that after a few years of high earnings growth, investors will assume that this will continue indefinitely and be prepared to overpay for such companies. This is particularly true for companies in cyclical sectors.

The Barber and Odean (2000) study found that whilst individual investors on average earned 3.7% per year less than risk adjusted market returns, the 20% most active investors earned about 10% less per year.

So how can individual investors significantly improve performance?

- Invest in ordinary shares for the long term and ensure that you have a 50-75% weighting in equities in your portfolio
- Do not monitor your portfolio daily as this may lead to regret and loss aversion decisions that are not in an investor's best interests
- Minimise trading to reduce transaction costs which are a drag on performance
- Do not sell the winners and retain the losers. Value each share at each price point in relation to future cash flows in order to reach buy or sell decisions
- Follow a disciplined investment strategy. Use an investment checklist in order to avoid impulse buying and selling
- Increase the number of shares to at least 15–20 shares in a portfolio in order to achieve diversification and avoid a home bias by investing in local and international companies.
- Do not simply apply a naïve *1*/*n* "diversification" strategy to asset allocation. If there are five available funds, then investors are inclined to invest 20% of their savings in each fund. This also means that you do not simply place 50% in equities and 50% in bonds. Asset allocation requires a deeper analysis of markets, instruments and risk
- On retirement, it is suggested that a retiree's asset allocation should be weighted towards low risk assets such as bonds. We consider this to be too simplistic when life expectancy for a retiree and the retiree's spouse may be another 25–30 years after retirement
- Reduce costs by investing in a low-cost index fund. Alternatively, invest in a unit trust fund if the fund has been able to outperform the index after fees over the long term

- Do not try and time the market. Historically, it has been best to remain invested rather than to try and time the market
- Take action. Do not let inertia stop you investing. Implement an investment strategy. Once you have invested in a diversified low-cost index, then inertia can be a positive force as compared to excessive trading. Automatic rebalancing will ensure that the portfolio retains its risk/return profile over time
- Be conscious of all biases such as overconfidence, conservatism, representativeness, anchoring and loss aversion as these can lead to a lack of diversification, overtrading, biased valuations and lack of stringent analysis and due diligence
- Avoid a narrow framing of investment decisions. A portfolio may be structured in a particular way to achieve diversification. If one sector experiences losses and values fall, then it is important to understand the hedging effects of such losses. For example, investing in exporting companies may result in losses if the Rand appreciates but the hedging objective remains valid and the local companies will perform strongly to offset these losses. Do not focus only on the losses in one section of your portfolio. We should evaluate portfolio performance in terms of its risk and return parameters rather than focus on the gains and losses of the individual shares that make up such a portfolio.

It is important to note also that traits such as inertia and procrastination have saved many investors, as well as cost them dearly. The Alexander Pope quote, *fools rush in where angels fear to tread* is pertinent here. However, Howard Marks when referring to bull markets, states that *what the wise man does at the beginning, the fool does at the end*. Slow deep analysis is different though and you may lose an opportunity but that is the price of caution. Overconfidence may project a sense of purpose and knowledge which can motivate others. Imagine listening to your doctor ponder and express doubts over your diagnosis[18]. However, confidence grounded on research and hard work is more valued than bluster.

Richards and Willows (2018) have studied the characteristics of investors that engage in frequent trading[19]. Dr Gizelle Willows has also studied the impact that gender and age have on investment performance in South Africa. It seems that women do better than men.

In Chapter 2, we found that not allocating a sufficient proportion of your portfolio to equities would endanger the longevity of your pension. Although equities implied higher volatility, not investing 50–75% in equities would mean that you would outlast your assets and end your days in financial hardship. It seems like the biggest risk you can take is not taking a risk.

APPENDIX 3.2 COMPOUND ANNUAL RETURNS, ARITHMETIC AVERAGES AND TOTAL SHAREHOLDER RETURNS

In this chapter we have determined the arithmetic average of an investment's returns and have computed the standard deviation to reflect the associated volatility and risk of an investment. The arithmetic average is simply summing the returns or values and dividing by the number of returns or values to determine an average. We have also used the compound annual growth rate (geometric average) to explain market returns. In this section we will explore arithmetic and geometric averages in greater detail before analysing the determination of total shareholder returns.

We will use an example, GEM Ltd, from Year 1 to Year 7 whereby the share price has risen with some volatility from R100 to R187. The compound annual growth rate is 11% (0.10996) per year.

Year	0	1	2	3	4	5	6
Price	100.0	140.0	121.0	170.0	110.0	130.0	187.0
Return		40.0%	-13.6%	40.5%	-35.3%	18.2%	43.8%

In Table 3.14, we have set out the workings of the above example in relation to the arithmetic average as well as the standard deviation of the returns. We have computed the compound annual growth rate (geometric mean) in three different ways and have set out the relevant Excel functions such as the =GEOMEAN function, which we apply to the range of 1+return numbers. Whilst Excel has a standard deviation function, we need to use logarithms in order to determine the geometric standard deviation factor which we apply to the geometric return factor (1+geometric return).

Table 3.14 Arithmetic means, geometric means and standard deviations

	A	B	C	D	E	F	G	H	I
4		Period	Price	Return	1+ return	Implied probability	100 x (1+GM)^t	Ln(1+return)	
5		0	100				100.00		
6		1	140	0.400	1.400	0.16667	111.00	0.34	=LN(E6)
7		2	121	-0.136	0.864	0.16667	123.20	-0.15	
8		3	170	0.405	1.405	0.16667	136.75	0.34	
9		4	110	-0.353	0.647	0.16667	151.78	-0.44	
10		5	130	0.182	1.182	0.16667	168.47	0.17	
11	N	6	187	0.438	1.438	0.16667	187.00	0.36	
12						1.00000			
13	Sum of returns			0.9365829	=SUM(D6:D11)	Arithmetic mean of Ln(1+return)		0.1043231	=AVERAGE(H6:H11)
14	**Arithmetic mean (average)**			**0.1560972**	=+D13/B11	1+geometric return		**1.1099590**	=EXP(H13)
15	**Arithmetic mean [Excel]**			**0.1560972**	=AVERAGE(D6:D11)				
16	Standard Deviation			0.3304802	=STDEV(D6:D11)	Standard Deviation of LN (1+return)		0.3269211	=STDEV(H6:H11)
17	Mean + 1 Standard Deviation			0.4865774		**GM SD factor**		1.3866920	=EXP(H16)
18	Mean - 1 Standard Deviation			-0.1743830		(1+geometric mean)*GM SD factor		1.5391713	
19	Ending value		187.0			(1+geometric mean)/GM SD factor		0.8004366	
20	Beginning value		100.0						
21	Ratio (End/Begin)		1.87		**Alternative**				
22	**Geometric mean**		**0.109959**	=+C21^(1/6)-1	(1.4*0.864*1.405*0.647*1.182*1.438)$^{(1/6)}$ =**1.10996**				
23	**Geometric mean [Excel]**		**0.109959**	=GEOMEAN(E6:E11)-1					

The arithmetic average of 15.6097% for GEM's returns is higher than the compound annual return (geometric mean) of 10.9959%.

The arithmetic mean will always be higher than the compound average growth rate (geometric mean) unless there is no volatility. As volatility increases, then the arithmetic mean will increase relative to the geometric mean.

Let's take another example and assume that the share price of HBL Ltd which was initially R100 goes up by 100% in Year 1 and falls by 50% in Year 2. The arithmetic average return is 25% [(100% + (–50%))/2]. The compound annual growth rate is 0%. In Year 1, the value went up by 100% to R200 and then fell by 50% back to R100.

When is the arithmetic mean relevant in this case? Well, if one was taking any gains off the table and putting back any losses, then the average return of 25% would be relevant as the Page 3-41 opening investment in Year 2 would be R100 and then you lose R50, so that at the end of the day you put back R50 to get back to your initial investment and you still have a R50 gain which represents an average return of 25%.

The arithmetic average return will always be higher than the geometric average return due to the asymmetry of returns. For example, if the share price of R100 falls by 20% in the first year and then rises by 20% in the second year, this will result in an arithmetic average of 0%, implying break-even and yet we would have lost money [R100 × (1–0.20) × (1.20) = R96]. The return in the second year will need to be higher in order to break-even. Due to the fall by 20%, we now have a base of R80 and therefore will need to achieve a return of 25% to achieve break-even [R100 × (1 – 0.20) × (1.25) = R100]. The compound annual return is $(R100/R100)^{1/2} - 1 = 0\%$. The arithmetic average needs to be 2.5% per year, in order to achieve the same result [(–0.20 +0.25)/2].

Outliers and volatility will tend to increase the arithmetic mean relative to a geometric mean and is less likely to reflect the data. Assume data points of R100, R90, R500 and R110. The average is R200 which is nowhere near most of the data points due to the outlier of R500.

It is also important not to let *the tail wag the dog*. Greater volatility may result in a higher arithmetic average without affecting compound annual returns. For example, the only difference between Company X and Y in Table 3.15 is that Company Y's price jumped in Year 2 and therefore its returns reflect greater volatility and variability than Company X. We are assuming zero dividends. We need to be cautious when comparing Company X to Company Y, that we do not conclude in this case that Company Y offers a higher average return because it is more volatile. In fact, it is because of its greater volatility, that the average return is higher. On a compound annual return basis there is no difference between Company X and Y.

Table 3.15 Volatility and average returns

Year	Company X		Company Y	
	Price	Return	Price	Return
0	100.00		100.00	
1	110.00	10.00%	180.00	80.00%
2	143.00	30.000%	143.00	-20.556%
3	133.10	-6.923%	133.10	-6.923%

Year		Company X		Company Y
	Price	Return	Price	Return
Sum of returns		33.077%		52.521%
Arithmetic average	1/3	11.026%	1/3	17.507%
Compound annual return		10.000%		10.000%

In general, when we are comparing companies over a long period, then arithmetic averages and compound annual returns will tend to be aligned with volatility and higher arithmetic average returns and compound annual returns will tend to reflect higher risk in the form of higher volatility. We see that this reflects the returns in the real world based on our analysis of historical financial returns in this chapter.

When we are analysing a long series of market data, then arithmetic averages can be useful but we need to be aware of the impact that volatility may have on arithmetic averages. Warren Buffett in our example reported a compounded annual return (geometric mean) of 19.93% as compared to 9.94% for the S&P500. If we calculate Buffett's arithmetic average return, then this is 24.09% as compared to 11.37% for the S&P500.

Geometric means (compound average growth rates) may also be significantly affected by beginning points and end points. If you can select your beginning and your ending, then this will affect your geometric mean (compound annual return). So, whilst arithmetic means are less accurate depending what happens in the middle of the data, geometric means can be affected by beginnings and endings.

We will assume that you are a fund manager and you need to report on your investment over a number of years. The index values and annual returns of your investment fund over the last six years is indicated below: Page 3-42

Year	0	1	2	3	4	5	6
Price	100	80	135	110	140	132	143
Return		-20%	69%	-19%	27%	-6%	8%

In terms of reflecting performance truthfully, your fund has achieved a compound annual return of 6.1% per year $[(143/100)^{1/6} - 1]$. This does not look that good. So, instead of reporting the performance of the fund over the prior six years, you select the beginning to be the end of Year 1, so that your return almost doubles to 12.3% per year $[(143/80)^{1/5} - 1]$. Of course, you can bolster your return further by reporting the arithmetic average of 16.0% over the five years from Year 2 to Year 6. These are all truths but may distort perceptions about future returns. In terms of framing returns, these may also be gross returns before fees which will reduce compound and average net returns. So, it is also relevant to watch how fees can affect net returns.

Total shareholder return

It is important to evaluate returns over time by considering the reinvestment of dividends. Returns are not only beholden to changes in prices but also to dividend distributions. Let's

compare investing in XYZ Ltd and ABC Ltd over the last six years. Table 3.16 depicts the share price of either company over this period of time.

Table 3.16 Share prices, annual returns and compound returns

XYZ Ltd	0	1	2	3	4	5	6
Price	**100.00**	70.00	120.00	135.00	110.00	140.00	**160.00**
Annual return		-30%	71%	13%	-19%	27%	14%
Compound annual return	**8.15%**	$[(160/100)^{1/6} -1]$					
ABC Ltd	**0**	**1**	**2**	**3**	**4**	**5**	**6**
Price	**100.00**	70.00	120.00	130.00	105.00	130.00	**145.00**
Annual return		-30%	71%	8%	-19%	24%	12%
Compound annual return	**6.39%**	$[(145/100)^{1/6} -1]$					

On the basis of price movements, XYZ has provided a compound annual return of 8.15% whilst ABC has only provided a compound annual return of 6.39% per year. However, ABC has been paying an annual dividend of R6 per share each year in Years 1, 2, 3 and 5 and paid a special dividend of R10 in Year 4. No dividend was paid in Year 6. How does this change our analysis of the relative performance of these two companies? It is important to compare not only price returns but total shareholder returns which includes the reinvestment of dividends.

In Table 3.17, we have recalculated the compound annual return of investing in ABC Ltd if a shareholder reinvests each dividend payable at the end of each year back into the company. At the end of Year 1, we use the R6.00 dividend to purchase 0.0857 (R6/R70) of one share. At the end of Year 2, we use the R6.00 to purchase 0.05 (R6/R120) of one share (Total: 1.0857 × 0.05 = 0.0543 shares). By Year 6, the reinvestment of the dividends will result in an investor holding on to 1.3665 shares by investing the dividends in shares so that the value of one share at a price of R100 will have grown to R198.14 (1.3665 × R145)

Table 3.17 Total shareholder return with the reinvestment of dividends

ABC Ltd	0	1	2	3	4	5	6
Dividend per share		6.00	6.00	6.00	10.00	6.00	-
Shares purchased with dividend#		0.0857	0.0543	0.0526	0.1136	0.0603	-
Cumulative shares	1.0000	1.0857	1.1400	1.1926	1.3062	1.3665	**1.3665**
Price per share	100.00	70.00	120.00	130.00	105.00	130.00	145.00
Value	**100.00**	76.00	136.80	155.04	137.15	177.64	**198.14**
End value/Beg value	1.9814			# DPS/Price x shares held prior to dividend			
Compound annual return	12.07%	$\Gamma(198.14/100)^{1/6}-1]$		# For example, Yr3: 6/130 x 1.14 = 0.0526 shares			

Page 3-43 The total compound annual return of investing in ABC Ltd, when one considers the reinvestment of dividends is 12.07%. This is much higher than the return for XYZ which has not paid a dividend. Therefore, consider the reinvestment of dividends when computing a total shareholder compound annual return.

Does this make a significant difference to equity returns over time in South Africa? It sure does. In fact, the difference is profound. Investing R100 in 1970, would have grown to R326 366 by 31 March 2023 if one had reinvested one's dividends. Without reinvestment, the cumulative value would be R37 595. Whilst this remains a good outcome, it is some way off the cumulative amount if an investor had reinvested all dividends.

The compound annual return from reinvesting dividends since 1970 has been 15.31% per year and 11.21% without the reinvestment of dividends. Why this is important, is that JSE returns will often disclose the total shareholder return (TSI) and you should not expect the compound annual return of the ALSI in the form of the TSI if you are spending your dividend. Be more modest in your expectations but if possible, also endeavour to reinvest your dividends as the power of compounding will lead to significantly higher returns over time. Dividends are much more important to equity returns than most people think. So be wise with dividends.

Also, consider the impact of fees on your returns. On average most actively, managed unit trusts or funds have not been able to outperform, after fees, the market index such as the ALSI or the S&P500 index in the USA. Whilst the current inflows may create long-term challenges, for now the momentum is with passive low-cost index fund investing. Even hedge funds have not been able to beat the market, after fees and I refer you to Berkshire Hathaway's annual report in which Warren Buffett reports on his bet against hedge fund performance in favour of a low-cost index fund investing strategy.

In this section, we have offered some insights into reporting on performance in the form of geometric averages (compound annual returns) and arithmetic averages. We have also reflected on the impact of the reinvestment of dividends on compound annual returns in specific examples and for the JSE.

SELF-STUDY PROBLEM

The following limited information is available for returns on two shares listed on the Stock Exchange.

Year	Gipley	Petros
20.1	0.20	0.16
20.2	0.28	0.12
20.3	0.36	0.10
20.4	0.12	0.18

Despite the limited number of readings, a normal distribution of returns may be assumed. In addition, past performance is considered to reflect expected future performance.

Required:

(a) Calculate the average return, standard deviation, and co-efficient of variation for each of the two shares.

(b) Using the properties of a normal distribution and the z score tables, establish the probability for each share of a return lower than 10%.

(c) Taking the role of an investment adviser, recommend one of the two shares to a client who wishes to make a choice between an investment in Gipley Ltd or an investment in Petros Ltd. Advise the client of some of the issues which should be considered. Page 3-44

Suggested solution

(a)

Year	r_s		dev R_g	dev R_p	dev R_g^2	dev R_p^2
20.1	0.20	0.16	-0.04	0.02	0.0016	0.0004
20.2	0.28	0.12	0.04	-0.02	0.0016	0.0004
20.3	0.36	0.10	0.12	-0.04	0.0144	0.0016
20.4	0.12	0.18	-0.12	0.04	0.0144	0.0016
	0.96	0.56			0.0320	0.0040
Mean	0,24	0.14		Variance	0.0080	0.0010
				Standard deviation	0.0894	0.0316

			Gipley	Petros
Average Return		a	24.00%	14.00%
Standard deviation		b	8.94%	3.16%
Co-efficient of variation		b/a	37.25%	22.57%

(b) Return lower than 10%

		Gipley	Petros
[Mean return - 10%]/σ	=	0.14 / 0.0894	0.04 / 0.0316
z	=	1.5660	1.2658
Table E [rounded z to G=1.57, P=1.27]		**0.4418**	**0.3980**
		0.5000	0.5000
Probability < 10%		0.0582	0.1020
Excel =NORMSDIST(z)	=	0.9413	0.8972
This is the probability that the return will be at least 10%.			
The probability that the return will be less than 10% is:			
		1.0000	1.0000
Probability => 10%		0.9413	0.8972
Probability < 10%[1]		**0.0587**	**0.1028**
This is the probability that the return will be less than 10%.			
1. Difference to prior result using Table E is due to rounding - Excel is accurate.			

(c) All investment choices depend on the individual investor's propensity for risk. It is apparent that Gipley is a more volatile share than Petros. Because of this characteristic, investors require a higher expected return than they do for an investment in Petros. This is evidenced from the higher return which has been achieved in the past.

Based on the analysis, there is still a lower probability of achieving a return of less than 10% by buying the shares of Gipley rather than Petros. However, investors who are very risk averse may consider even such a risk to be unacceptable. It can be calculated that the probability of suffering a negative return from an investment is around 0.5% or one chance in 200, while there is virtually no chance at all, based on probabilities, of suffering a negative return from an investment in Petros.

The most significant issues, once probabilities based on past performance have been discussed, relate to future expectations. We are not told, for example whether these companies are both listed in the same sector. Industry factors and the investor's expectations regarding the future would be considered. Page 3-45

QUESTIONS

Question 3.1

A dictionary defines risk as "the chance of bad consequences", and uncertainty as "not to be depended on".

Required:

(a) Discuss these terms in the context of financial management.

(b) Identify the relationship between estimated risk and expected return, with respect to a financial investment.

(c) Explain why an investment in a government security may be considered to be a riskless investment.

Question 3.2

When you take a risk you should be rewarded with a return. Why is this nexus the cornerstone of finance?

Question 3.3

Different financial instruments have different inherent risks. Describe the risks associated with the following:

1. Government bonds	5. Preference shares
2. Corporate bonds	6. Convertible notes
3. Debentures	7. Convertible Preference shares
4. Ordinary shares	

Question 3.4

When assessing the overall risk of a company, financial managers take cognizance of business risk and financial risk.

Required:

(a) Define business risk and discuss the method by which it may be measured.

(b) Define financial risk and discuss the method by which it may be measured.

(c) Discuss the actions a financial manager may take to reduce the overall risk of the company and the effects this may have on shareholders.

Question 3.5

The following information is available for Astrid Ltd and Duncast Ltd:

	Astrid	Duncast
Units produced and sold	25,000	25,000
	Rm	**Rm**
Revenues	112.5	112.5
	75.0	75.0
Variable costs	25.0	50.0
Fixed costs	50.0	25.0
Net Operating income	37.5	37.5

Required:

(a) Find the break-even point for each company in units.

(b) Calculate the degree of operating leverage for each company at 25 000 units. Page 3-46

(c) Explain the differences that you observe between these companies' break-even points and degrees of operating leverage.

Question 3.6

A small company, Flexet (Pty) Ltd has reported the following results for the past year:

Inconie Statement for the year ended 31 December	
Sales (30000 units at R250)	7,500,000
Variable costs (30000 units at R70)	2,100,000
Contribution	5,400,000
Fixed costs	2,700,000
Earnings before interest and tax	2,700,000
Interest expense	1,700,000
Earnings before tax	1,000,000
Taxation	280,000
Earnings after tax	720,000

Required:

(a) Calculate the degree of operating leverage.

(b) Calculate the degree of financial leverage.

(c) Calculate the degree of combined leverage.

(d) Find the break-even point in units.

Question 3.7

The following are abbreviated contribution income statements of two companies, both of which are active in the food and beverage industry.

	Quickpro Rm	Hendels Rm
Sales	290	290
Variable costs	58	145
Contribution	232	145
Fixed costs	120	75
Earnings before interest and tax	112	70
Interest expense	60	18
Net income	52	52

Required:

(a) Calculate the degree of operating leverage.

(b) Calculate the degree of financial leverage.

(c) Calculate the degree of combined leverage.

(d) Find the break-even point in rands.

(e) Discuss the relative riskiness of the two companies based on the leverage factors calculated.

Question 3.8

An investor purchased 500 shares in Glicks Stores Limited at the beginning of its financial year at R22.50 per share, paying 3% brokerage costs. The book value per share on that date was R10.60. During the course of the year, the company paid a dividend of R2.40. One year later, the company reported earnings per share of 470 cents. On that date the book value per share was reported as R13.80. The investor sold the shares for R30.70 each, paying 3% brokerage on the transaction. Page 3-47

Required:

(a) Calculate the return which the shareholder earned for her investment in Glicks stores Ltd.

(b) Discuss the difference between return on equity, return on investment, and the return to the shareholder.

(c) If the investor had decided not to sell the shares at the end of the year, calculate her return.

(d) Comment on the riskiness of the investment in Glicks Stores Ltd based on the information provided.

(e) If the period had been eight months rather than one year, all other information remaining the same, calculate the return to the shareholder.

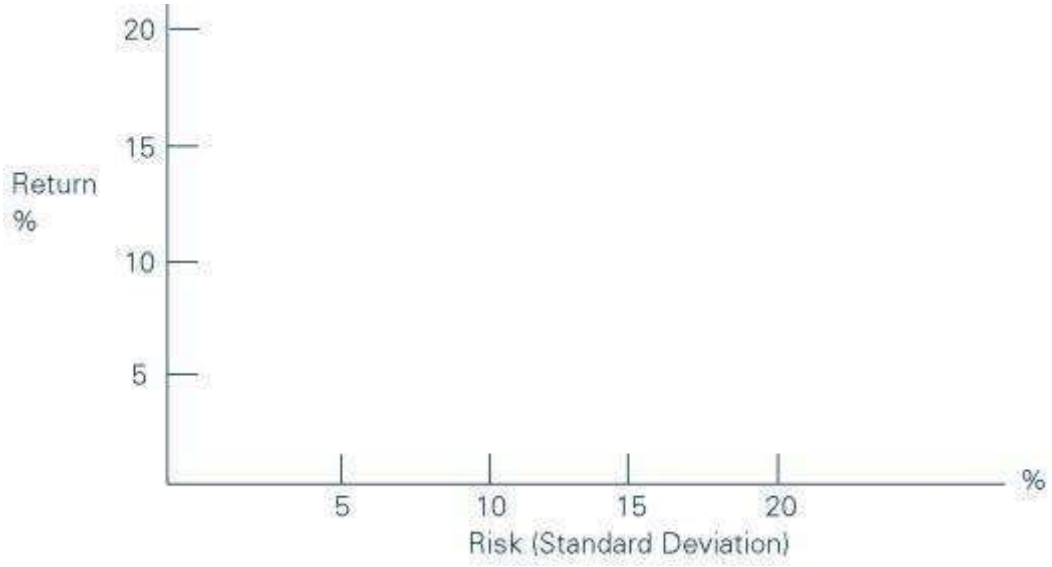
Question 3.9

You wish to purchase shares in two companies of the following six that are available:

- Company A with an expected return of 10% with a standard deviation of 10%
- Company B with an expected return of 10% with a standard deviation of 15%
- Company C with an expected return of 15% with a standard deviation of 15%
- Company D with an expected return of 10% with a standard deviation of 20%
- Company E with an expected return of 15% with a standard deviation of 20%
- Company F with an expected return of 20% with a standard deviation of 20%

Required:

(a) Plot these shares on a risk/return graph such as indicated below.



(b) As an investor who is willing to take risks in the hope of retiring early, which **two** shares would you buy? Explain.

(c) As a cautious investor who requires a steady return on your investments, which **two** shares would you buy? Explain.

(d) Investors are said to be risk averse. Does this mean they will not invest in risky assets?

(e) Why are government bonds considered low risk when there is always the chance that interest rates will change and the value of the bond will change accordingly?

Question 3.10

Lighthouse Ltd is considering investing in a project that may offer the following possible rates of returns. The probability distribution of returns from the project is set out below:

Probability	10%	20%	40%	20%	10%
Return	-30%	-5%	15%	30%	40%

What is the expected return from the project? What is the standard deviation of returns? Page 3-48

Question 3.11

Emma Dlamini is considering two investments and can only invest in either the shares of Company X or in the shares of Company Y. The following information regarding returns and probability distributions of returns is relevant:

Share X		Share Y	
Probability	Return X	Probability	Return Y
0.20	10%	0.20	0%
0.60	15%	0.30	10%
0.20	20%	0.30	20%
		0.20	30%

Required:

(a) What is the expected return of X and Y?

(b) Calculate the standard deviations of X and Y's returns

(c) What is the co-efficient of variation?

(d) What would you recommend? Why?

(e) Instead of Share Y offering a probability of 20% of achieving a 30% return, assume instead that the possible return is 40%. How would this affect the investment decision?

Question 3.12

Home Wares Ltd is a listed company which provides home solutions. The company has benefited indirectly from the growth in residential developments. As an investor in the shares of the company, you wish to determine how the company's shares have performed in relation to the market. You would like to also understand the relative risk of investing in the company as compared to the market. The following information on returns for Home Wares and the market is relevant:

Year	Home Wares	Market
1	12%	8%
2	-6%	-4%
3	-9%	-6%
4	15%	10%
5	18%	12%

Determine the average return for the company and the market. What is the standard deviation for the company and the market? Comment on the results.

Question 3.13

Based on various scenario planning and economic forecasts, the probability of various states of the economy have been determined by an economic analyst. The various possible states of the economy depend upon factors such as the national debt, foreign investment prospects and internal political factors. The financial manager of Plus-Two Ltd, manufacturers of fashion wear for teenagers, have predicted the most likely return to shareholders given the state of the economy as follows:

Conditional state of the economy	Probability of state of economy	Expected return
Boom	5%	40%
Strong economy	15%	20%
Moderate economy	60%	10%
Recession	15%	0%
Deep recession	5%	-10%

Page 3-49 *Required:*

(All calculations may be rounded to two decimal places of a percentage.)

(a) Calculate the expected return.

(b) Calculate the variance of the expected return.

(c) Calculate the standard deviation of the expected return.

(d) Assuming a normal distribution of returns, calculate the probability that shareholders will receive a return in excess of 25%.

Question 3.14

Environ Ltd has gathered the following information regarding the probable costs of two alternative projects designed to recycle effluent which is environmentally sensitive. One of the projects must be accepted in accordance with legislation relating to environmental pollution. The possible net cash flows have all been discounted at 16%, but are heavily dependent upon factors that cannot be predicted with certainty. However, probabilities of occurrence have been established.

Project A - Liquid Disposal	
Net Cash Flows (PV)	Probability
Rm	
-150	20%
-85	50%
-10	10%
50	20%

Project B - Solid Disposal	
Net Cash Flows (PV)	Probability
Rm	
-120	30%
-40	40%
0	20%
50	10%

Required:

(a) Calculate the expected present value (or cost) of each of the two projects.

(b) Using mean–variance analysis, and the assumption of continuous probabilities, recommend which project should be accepted.

(c) Discuss the issues that will require consideration depending upon whether the probabilities are discrete probabilities or a continuous probability distribution.

Question 3.15

Siyabonga Ltd is attempting to measure the riskiness of two projects which have the following cash flows in different, equally likely, states of the economy.

State of Economy	State 1	State 2	State 3	State 4	State 5
Project G	900	800	400	300	200
Project H	900	500	400	300	100

Required:

(a) Establish, using the standard deviation of cash flows as a measure of risk, which project is riskier.

(b) Establish, using the co-efficient of variation of cash flows as a measure of risk, which project is riskier.

(c) Explain which measure of risk you would prefer.

Question 3.16

Baxter Ltd is faced with a problem of fluctuating costs of materials which differ, dependent upon the type of machine which is used. Two different machines, each producing the same product, are available. Various possible fluctuations from the estimates have been established for each machine. The resultant table of possible costs and probabilities assigned to those costs is as follows: Page 3-50

Flexing machine		Binding machine	
Probability %	Costs R000	Probability %	Costs R000
10%	750	20%	1,050
40%	900	30%	1,400
30%	1,420	50%	1,300
20%	1,500		

Required:

(a) Determine the most likely cost for each machine, assuming:

(i) that the probabilities are discrete;

(ii) that the probabilities reflect point estimates on a continuous distribution which closely approximates a normal distribution.

(b) For each machine, determine the probability that the cost will be in excess of R1.2 million, using the properties of a normal distribution and z scores.

Question 3.17

The past returns on an investment in the shares of Nwabisa Ltd have been recorded and processed to produce the following statistical indicators:

Mean return	24%

Standard deviation	9%

You may assume that the distribution around the mean has the properties of a normal distribution.

Required:

(a) State the most likely return which an investor may expect to receive from an investment in the shares of Nwabisa Ltd.

(b) Discuss the issues which should be considered in establishing the likelihood of the return being different to the expected return.

(c) Using z scores, establish the probability of the following outcomes:

(i) a return of more than 33%

(ii) a return of less than 10%

(iii) a negative return.

Question 3.18

The following summary statistics are generated for the investments listed below.

	Mean return	Standard deviation
Government bonds	8%	2%
Paveco Ltd	14%	8%
Energet Ltd	18%	12%
Argyle Ltd	20%	11%

You may assume that the distribution around the mean has the properties of a normal distribution.

Required:

(a) Provide reasons (if such reasons exist) as to why an investment in a government bond may have a standard deviation other than zero.

(b) Calculate the co-efficient of variation for each of the three shares.

(c) Assume an investor is considering purchasing one of the above four shares. Identify considerations which should be taken into account and make a recommendation, if possible. Page 3-51

(d) Identify an incongruity which exists between the summary statistics of Energet and Argyle and discuss whether such summary statistics are feasible.

(e) Calculate the probability of the return on investment in the shares of Paveco being negative.

Question 3.19

As a financial planner you are called upon to advise your clients on the shares they should invest in. The choices are:

Company	A	B	C	D	E
Return	10%	10%	15%	10%	20%
Standard deviation	10%	15%	15%	20%	20%

(a) Mr Fossil who is risk averse wants you to rank the shares in the order he should consider investing in them.

(b) Mr Steady is risk neutral and also seeks the appropriate order of investments in which he should invest.

(c) Mr Quickbuck is a risk taker who wants to retire early in life. Suggest the appropriate order of the above investments that will satisfy his risk profile.

Question 3.20

The following returns have been obtained from the company's respective financial statements:

Company	20x1	20x2	20x3	20x4	20x5	20x6	20x7	20x8	20x9
S Ltd	18%	15%	15%	18%	18%	5%	-10%	18%	20%
T Ltd	10%	12%	14%	12%	12%	12%	12%	10%	12%

(a) Calculate the expected return and standard deviation of the above companies.

(b) An investor requires high returns but is indifferent as to the risk. Which company would he invest in?

(c) You are a risk averse investor. Does your choice change?

Question 3.21

SQ Ltd is listed on the JSE and its share price at end of each year is set out as follows:

SQ Ltd	20x0	20xl	20x2	20x3	20x4
Share price	50.00	75.00	60.00	80.00	100.00

SQ paid no dividend in 20x1, but consistently paid a dividend of R8 per share at year end in each year from 20x2 to 20x4. Prices are stated ex-dividend.

Required:

(a) Determine the arithmetic average annual return and compound annual return for the period until 20x4, based on price movements only and ignoring dividends.

(b) What is the arithmetic average return including annual dividends?

(c) What is the **total shareholder return** (compound annual return) if dividends are reinvested at the end of each year?

[1] *Credit Suisse* Global Investment Returns Yearbook, 2009.

[2] The MSCI World Index reflects the performance of large and mid-cap companies in Developed Markets.

[3] Minsky was a US economist who studied how excessive borrowings and prosperity can lead to financial instability and a Minsky moment refers to a sudden sharp decline in markets after a period of economic prosperity, excessive borrowings, low volatility and high asset valuations.

[4] If we apply =NORMSINV(0.95) in Excel for a 95% confidence level, then the answer is 1.6449 standard deviations. Otherwise, apply =NORM.INV(0.05, 0.00742,0.0639) to determine VaR of -9.769%.

[5] The distance between -10.282% and -10.252% = 0.030% and we multiply this by 25% to get to 0.0075% and then we add this to – 10.282% to get to an approximate value of –10.274%. More accurately, we should use 5% ×426 [n+1] = 21.30 and 0.30 x 0.030% = 0.009% so that -10.282% + 0.009% = -10.273%. We can also use the Excel function =PERCENTILE.EXC(F38:F462,0.05) where you specify the data range (in our case this was F38 to F462) and the 5th percentile. Excel will indicate the return below which 5% of the returns are. Excel calculates this to be –10.273% for emerging markets and –7.712% for developed markets.

[6] Kurtosis measures the likelihood of extreme values from the mean and therefore the likelihood of fat tails. A Kurtosis score above zero (excess Kurtosis) will indicate fatter tails than indicated by a normal distribution. If we employ the Excel function, =KURT, to our data set (LN returns), we find that the Kurtosis is 3.203 which indicates the existence of fat tails. [Please note that the Excel function = KURT refers to excess kurtosis.]

[7] The use of the term *compound annual growth rate (CAGR)* has become a standard way of describing a geometric average. It may not need to refer to an annual period and we will also refer to the terms "*compound average growth rate*" and "*compound annual return*" to depict a geometric average.

[8] The product of the annual returns is: 1.17*1.3909*0.9308 = 1.5147 which gives the same 51.47% HPR. Thanks to Hendrik Wolmarans of UP for pointing this out and for correcting a prior version.

[9] How did we work this out?

Equity: $FV = R1 \times (1.07)^{123} = R4113.4$

Bonds: $FV = R1 \times (1.018)^{123} = R9.0$

Please note that there are slight rounding differences in terms of the *Credit Suisse* returns indicated in Table 3.8.

[10] See Auret and Cline (2010). *Size and value investment strategies on the JSE*, Conference Paper presented at the African Finance Journal Conference, Stellenbosch, 2010.

[11] *Credit Suisse* Global Investment Returns Yearbook, 2010. In 2023, Bekaert, Harvey & Mondino write in *Emerging Markets Review* (https://doi.org/10.1016/j.ememar.2023.101034) that "Betas now seem to fluctuate within a 1.1-1.4 band, making emerging markets a risky, high-expected return asset class." Betas are explained in greater detail in Chapter 4.

[12] An investment in shares from the rest of the world would only have reached a value of $176 which reflects the dominance of the USA economy and equity markets over the period, but still much better than gold.

[13] Information in these paragraphs is sourced from the *Credit Suisse Global Investment Returns Yearbook, 2023 Summary Edition*, by Elroy Dimson, Paul Marsh and Mike Staunton.

[14] JP Morgan *The agony and ecstasy: the risks and rewards of a concentrated stock position*, in Eye on the Market: Special Ed., 2014.

[15] We set out an overview of behavioural finance in Chapter 1 in Appendix 1.1. The definitions of biases referred to in this section are set out in Chapter 1 Appendix 1.1.

[16] Barber, B.M. and Odean, T. (2000), Trading is hazardous to your wealth: The Common Stock Investment Performance of Individual Investors, *Journal of Finance*, 55, 773–806.

[17] Apparently, Fidelity found that the best investors were either dead or had forgotten they had an investment with Fidelity. Inactivity pays as investors do not get in the way. The Fundsmith Equity Fund's approach is "buy good companies, don't overpay and do nothing". Whilst finding good companies is challenging, most investors find that doing nothing is also hard to do.

[18] Studies have found whilst doctors were very confident of their diagnoses whilst patients were alive, it was found that up to 40% of such diagnoses were wrong when these patients were subject to autopsies after death. It gets worse. Dr Michael Greger in his book, *How not to die*, lists medical care as the third leading cause of death in the USA.

[19] Richards, D.W. and Willows, G.D. (2018), Who trades profusely? The characteristics of individual investors who trade frequently. *Global Finance Journal*, 35, 1–11.